

Deep peptide recognition profiling decodes TCR specificity and enables disease-associated antigen discovery

Received: 14 July 2025

Accepted: 13 April 2026

Published online: 13 May 2026



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Predicting T cell receptor (TCR) specificity on the basis of sequence is challenging because TCRs of similar sequence can recognize entirely different antigens, whereas TCRs of different sequence can recognize the same antigens. Here we present a system that integrates high-throughput yeast display with fine-tuned protein language models (pLMs) to generate deep peptide recognition profiles (PRPs) for individual TCRs, each detailing binding against millions of peptides. We provide detailed PRPs for a panel of HLA-B*27:05-restricted TCRs from persons with ankylosing spondylitis and acute anterior uveitis that almost exclusively recognize peptides through CDR3 β . pLMs trained on these PRPs outperform AlphaFold3 and tFold-TCR in predicting T cell activation. We discover and validate novel candidate autoantigens, demonstrate that model generalization to new TCRs correlates with functional distance (PRP divergence) rather than sequence similarity and introduce a model-intrinsic uncertainty metric to quantify prediction confidence. This system and its associated PRP datasets offer a scalable approach to mapping TCR recognition, accelerating antigen discovery and guiding TCR engineering.

T cell receptors (TCRs) recognize a composite surface composed of antigenic peptide and major histocompatibility complex (pMHC) molecules¹. Interrogation of pMHC complexes by the TCR underpins adaptive immunity, orchestrating responses to pathogens, cancer and autoimmunity^{2,3}. However, the structural complexity of the TCR–pMHC interface presents both challenges and opportunities for biotechnological intervention⁴. A fundamental paradox of TCR recognition is that TCRs with little to no sequence similarity can bind the same

pMHC, while nearly identical TCRs can exhibit distinct specificities^{5,6}. Consequently, sequence-based clustering methods such as GLIPH and TCRdist capture broad statistical trends in epitope specificity^{5,7}. However, they are limited in resolving fine-grained specificity among closely related TCRs.

Current approaches to bridge this sequence–function gap face notable limitations^{8–10}. While high-throughput experimental techniques such as yeast or mammalian display can map peptide

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interactions at scale, leveraging these large datasets effectively to build predictive models remains challenging^{11–16}. For example, computational approaches, whether based on TCR sequence similarity heuristics or structural modeling as exemplified by AlphaFold3 (ref. 17), a general structure-prediction model, and tFold-TCR¹⁸, a specific TCR–pMHC structure-prediction model, both of which use the interface predicted template modeling score (ipTM) to rank interactions, often struggle to reliably predict *de novo* TCR binding with high accuracy, particularly when confronted with unseen epitopes and diverse TCR backgrounds^{19–23}. Furthermore, synthetic peptide libraries used in display platforms frequently lack direct relevance to the native proteome of interest. This predictive shortfall hinders progress in critical areas, such as the efficient identification of elusive autoantigens driving HLA-associated autoimmune diseases, such as ankylosing spondylitis (AS) and acute anterior uveitis (AAU), two overlapping syndromes linked to HLA-B*27.

Here, we address these challenges with an integrated experimental–computational platform that generates deep peptide recognition profiles (PRPs) for TCRs whose recognition of pMHC has been visualized at high resolution. Our approach couples high-throughput yeast display, screening individual TCRs against millions of peptides, with protein language models (pLMs) fine-tuned on the resulting empirical binding data. The extreme focus of CDR3 β on peptide recognition in this TCR family²⁴, as determined from high resolution crystal structures, enables us to map and rationalize structure–activity relationships with an unusual degree of granularity. This PRP data resource reveals that disease-associated TCRs cluster by shared ligand recognition, complementing but extending beyond sequence-based grouping approaches. Critically, pLMs trained on these PRPs predict functional T cell activation with much greater accuracy than existing structural modeling approaches. This predictive power directly enabled the identification of novel candidate autoantigens for AS and AAU, including a peptide derived from PSG5, which we validated in individual-specific T cells. Lastly, we establish the principles governing model generalization, demonstrating that predictive accuracy for a new TCR is determined by its functional distance, not sequence similarity, to the training set and we introduce a model-intrinsic metric to quantify this confidence.

Together, this work highlights how integrating deep profiling with machine learning can ‘zoom in’ on narrow, disease-associated TCR families, revealing specificity rules not accessible by sequence-based methods alone. This platform and its associated resources provide a complementary path toward antigen discovery, deepen our understanding of immune specificity and guide the development of next-generation engineered immunotherapies.

Results

An integrated platform maps TCR–peptide recognition landscapes

To resolve the complex relationship between TCR sequence and peptide antigen specificity, we developed an integrated experimental–computational platform that empirically maps the peptide recognition landscape of individual TCRs (Fig. 1). The experimental framework uses high-throughput yeast surface display coupled with next-generation deep sequencing to quantitatively measure the binding interactions between a specific TCR and vast, randomized peptide libraries presented on the MHC (Extended Data Fig. 1). To generate high-resolution data for machine learning, we designed the peptide libraries to maximize informative diversity within the canonical binding space of our target system, HLA-B*27:05. Building on prior observations that disease-associated TCRs strictly prefer proline at position 8 (P8), we fixed the anchor residues (arginine at P2, proline at P8) while leaving the central positions and P9 unconstrained. This design trade-off ensured library stability and deep coverage of the variable contact residues most critical for fine specificity.

We applied this framework to a panel of 16 HLA-B*27:05-restricted TCRs implicated in AS and AAU, using this defined cohort as a reductionist model system for high-resolution mapping. This approach generated deep peptide recognition datasets, counting specific interactions for thousands of unique peptide ligands ($\sim 10^9$ peptides screened per TCR; Fig. 1a and Extended Data Fig. 2). These dense, quantitative PRPs provide a detailed view of each TCR's intrinsic peptide recognition preferences.

We reasoned that these rich interaction data could enable pLMs to learn generalizable rules of recognition. Accordingly, we developed a computational framework centered on pLMs, fine-tuned on these interaction datasets, to learn the complex sequence-dependent relationships governing TCR–pMHC binding (Fig. 1b). For the HLA-B*27:05 system studied here, we focused the models on the CDR3 β loop, which prior structural analyses identified as the dominant mediator of peptide contact in this system^{24,25}. This integrated platform enables the direct modeling of TCR specificity based on empirical peptide binding, allowing exploration of TCR relationships defined by shared ligand recognition patterns rather than TCR sequence similarity alone (Fig. 1c).

PRPs reveal functional clustering distinct from TCR sequence similarity

The 16 disease-derived TCRs encompassed the major disease-associated clonotypes previously described in synovial, blood and ocular compartments, including the dominant public TRAV21–TRBV9 family with the Y/FSTDITQ-BJ2.3 motif, a non-BV9 clonotype (O19.1) isolated from AAU aqueous humor that pairs TRAV21 with an alternative β germline, and additional clonotypes identified by YeiH tetramer sorting from blood from an individual with AAU that include both TRBV9 and TRBV5 usage^{24,26}. Alignment of these receptors reveals discrete CDR3 sequence variation within a clonotypically constrained family (Fig. 2a).

Structurally, available crystal complexes of representative HLA-B27:05-restricted AS and AAU TCRs reveal a recurring β -centric docking geometry²⁴. In all complexes, the CDR3 β loop arches over the center of the HLA-B27:05-bound peptide and accounts for most peptide backbone and side-chain contacts, whereas CDR3 α is positioned more peripherally. CDR1 α does contact the peptide but adopts a highly conserved configuration across TCRs (Fig. 2b). In this architecture, variability in CDR3 α is, therefore, most consistent with neutral drift around a fixed germline scaffold, while sequence differences in CDR3 β are directly translated into altered peptide contacts. This β -dominated binding geometry provides the mechanistic rationale for focusing our modeling on CDR3 β in this disease-linked TCR family.

Analysis of the experimental binding landscapes for this panel revealed substantial differences in the breadth of peptide recognition, with the number of unique peptide ligands identified per TCR ranging from hundreds to over 6,000 (Fig. 2c). To systematically compare TCRs on the basis of their recognition patterns, we calculated the pairwise Jensen–Shannon (JS) divergence between the peptide enrichment probability distributions derived from the yeast display data (Extended Data Fig. 3a). This functional distance metric (reflecting PRP divergence) revealed distinct clusters of TCRs sharing similar ligand preferences. Indeed, this clustering based on peptide recognition was largely independent of sequence similarity, as measured by CDR3 β edit distance (Extended Data Fig. 3b).

Visualization using multidimensional scaling (MDS) projected the TCRs into a peptide recognition space where proximity directly reflects shared peptide specificity, clearly delineating these recognition-based clusters and outliers (Fig. 2d, left). For instance, TCRs 135.1, 135.3 and 135.8 formed a tight peptide recognition cluster despite sequence differences, while sequence-similar TCRs sometimes exhibited disparate binding profiles. Correspondingly, sequence logo analysis of enriched peptides for each TCR identified distinct binding motifs (Extended Data Fig. 2). While some TCRs shared related motifs (for example, 28.1 and 135.4 both strongly favoring a

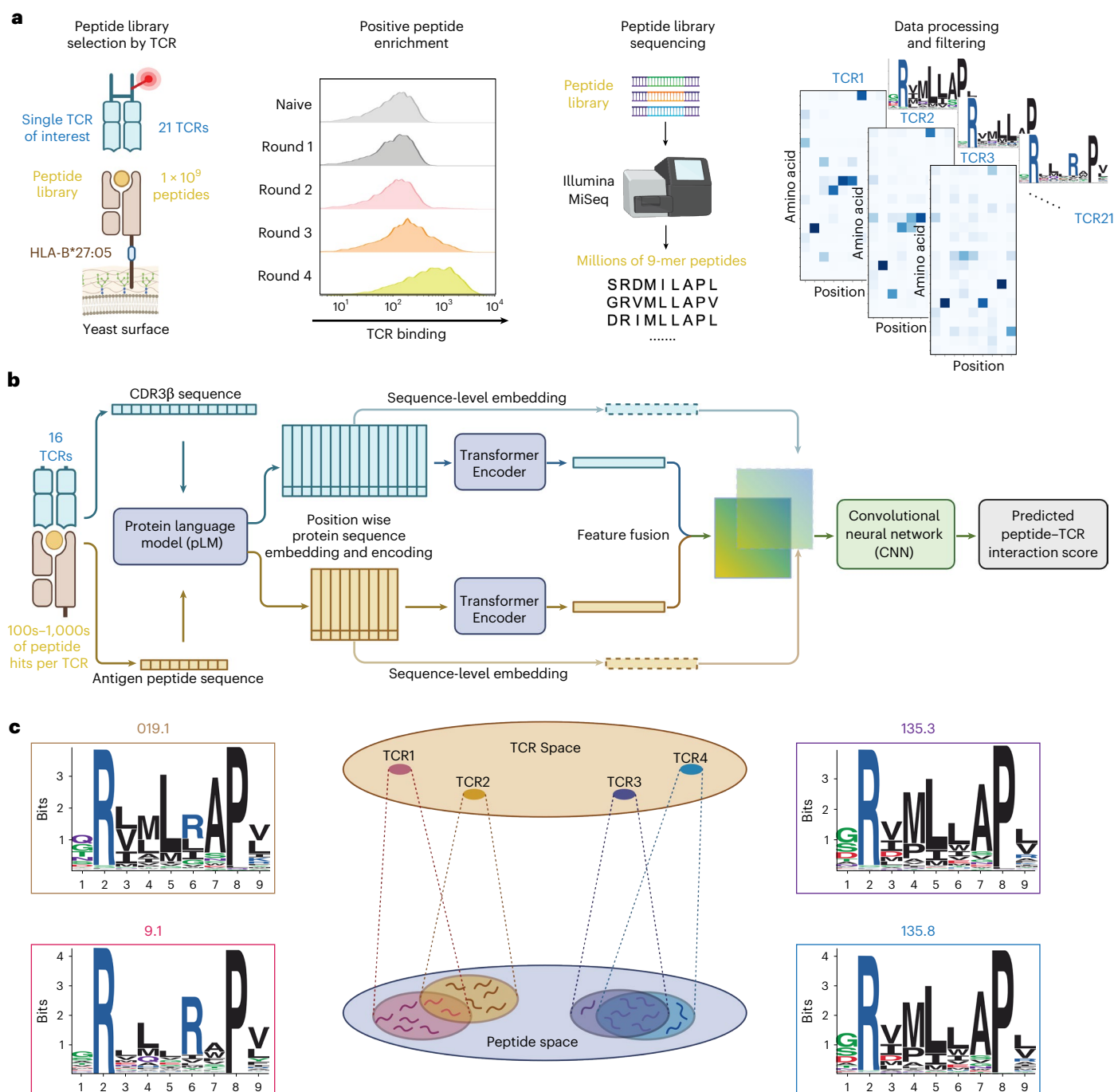


Fig. 1 | Integrated experimental and computational framework for modeling TCR–peptide binding specificity. **a**, Overview of the peptide screening workflow. Yeast display selections were performed using a 9-mer peptide library ($\sim 10^9$ variants) and 21 HLA-B*27:05-restricted TCRs (16 disease-derived TCRs used for modeling plus five engineered variants of clonotype 19.2 used later for mutagenesis validation in Fig. 5). Enriched peptides across four selection rounds were sequenced and processed to derive binding motifs for each TCR. **b**, Schematic of the machine learning pipeline. Representations of CDR3 β and

peptide sequences are computed from a fine-tuned pLM, followed by sequence-level and pairwise encodings, feature fusion and prediction of TCR–peptide binding scores. **c**, Conceptual model of TCR specificity landscapes as a two-layer manifold. Individual TCRs (top layer) define distinct yet overlapping PRPs (bottom layer), mapping the relationship between TCR sequence space and peptide specificity space. Representative peptide logos are shown for selected TCRs. Images in **a–c** were created in BioRender; Wang, N. <https://BioRender.com/8bgzti5> (2026).

central MLL motif at positions 4–6), others displayed unique preferences (for example, 9.1 with a unique preference for arginine at P6), consistent with their positions in the MDS map of peptide recognition space (Fig. 2d, right). Across dimensionality reduction methods (uniform manifold approximation and projection (UMAP), *t*-distributed stochastic neighbor embedding (t-SNE) and principal component analysis (PCA)), the same subsets of TCRs consistently clustered

together, mirroring their JS distance relationships and underscoring the robustness of the recognition space representation (Extended Data Fig. 3c,d). These results empirically demonstrate that comprehensive PRPs provide a more robust and biologically relevant measure of TCR relatedness than CDR3 β sequence similarity alone, directly revealing convergent recognition patterns among sequence-dissimilar receptors.

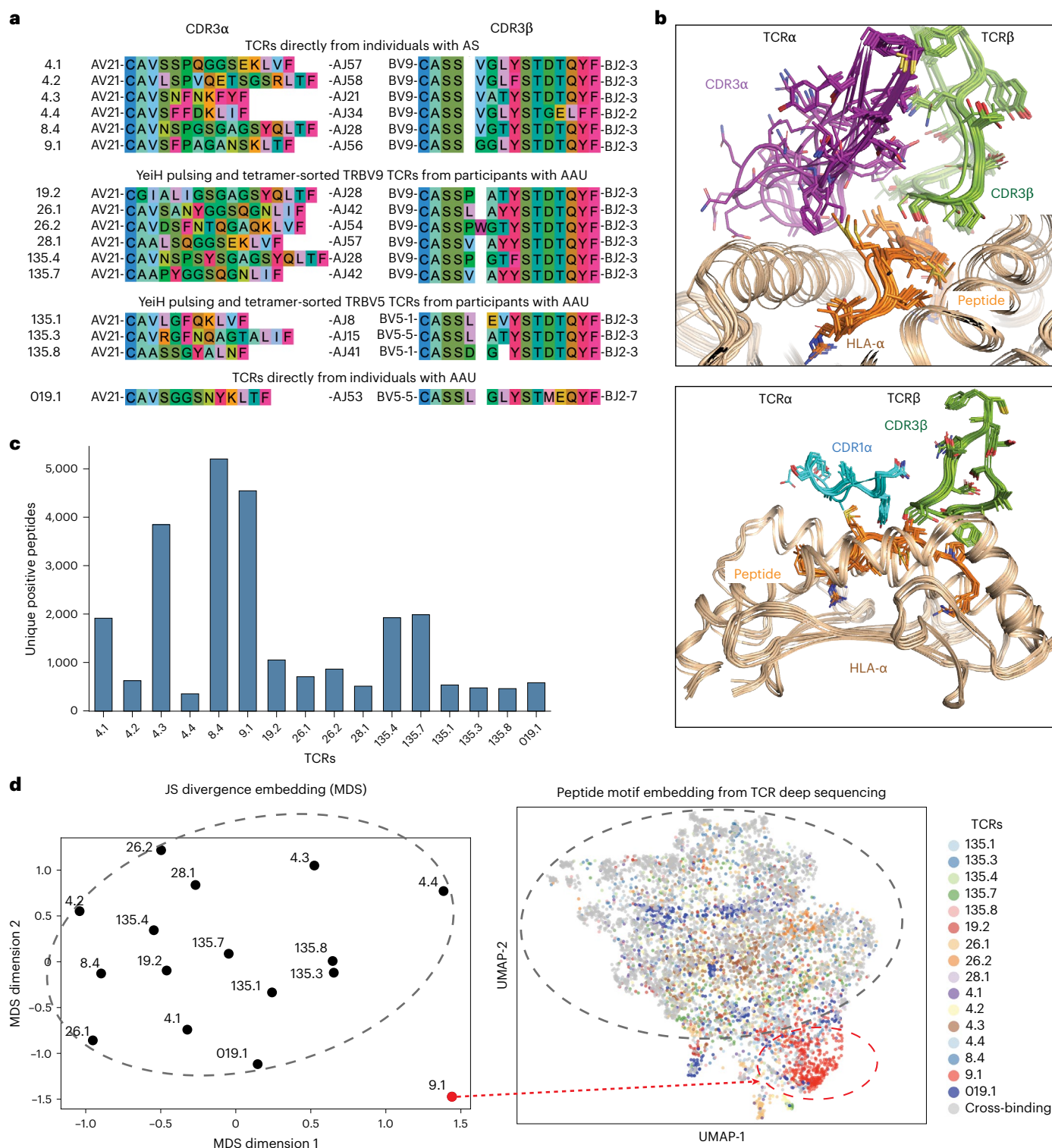


Fig. 2 | Functional diversity of HLA-B*27:05-associated TCRs in AS and AAU.

a, CDR3 α and CDR3 β sequence alignments of representative TCRs, grouped by source and isolation strategy. TCR identifiers are shown on the left.

b, Structural overlay of HLA-B27:05-restricted TCRs from individuals (PDB 7N2N, 7N2Q, 7N2P, 7N2R, 7N2S and 8CX4), illustrating the β -centric docking geometry. CDR3 β (green), CDR1 α (cyan) and CDR3 α (purple) are shown relative

to peptide (orange) and HLA-B27:05 α -helix (tan). **c**, Number of unique peptide binders identified for each TCR using high-throughput yeast display and deep sequencing. (criteria defining positive binding in the Methods) **d**, Left, MDS plot based on JS divergence of PRPs. Right, UMAP projection of enriched peptides from all TCRs based on their sequence embeddings. Each point represents a peptide, colored by the TCR(s) it binds to.

PRP-informed pLM accurately predict peptide-binding specificity

To translate the rich PRP datasets into a predictive tool, we fine-tuned pLMs for each TCR to learn its specific sequence–function map

(Fig. 3a). These models accurately discriminated binding from non-binding peptides in held-out test sets from the yeast display experiments, achieving high performance across the diverse TCR panel (area under the receiver operating characteristic curve (AUROC) > 0.95,

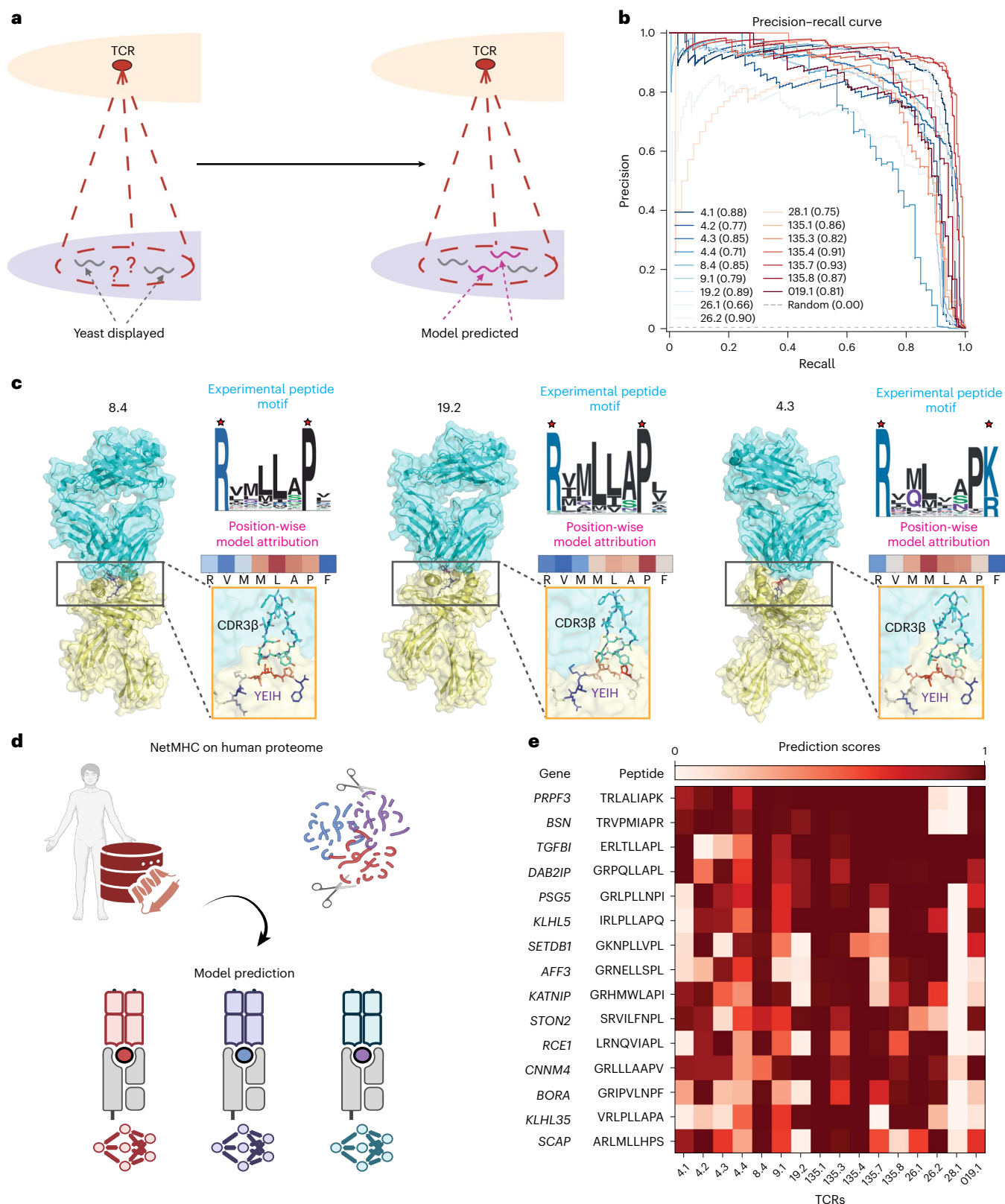


Fig. 3 | Interpretable machine learning models predict TCR–peptide interactions and enable proteome-wide antigen discovery. **a**, Schematic illustrating inferring a TCR's binding repertoire from partial experimental data using a trained model. Images in **a,d** were created in BioRender; Wang, N. <https://biorender.com/uvqm2r7> (2026). **b**, Model performance per TCRs shown as PR curves. Numbers in parentheses indicate the AUC. **c**, Structural analysis of peptide-binding preferences for TCRs (cyan) 8.4, 19.2 and 4.3 binding HLA-B*27:05 (yellow). Zoomed-in views highlight the CDR3β–peptide interface.

Peptide residues are colored according to gradient-derived positional importance (blue to red), with corresponding experimental binding motifs shown above. Red stars indicate fixed peptide positions. **d**, Workflow for nominating candidate autoimmune peptides. The 9-mer peptides predicted to bind HLA-B*27:05 were scored using individual TCR models. **e**, Heat map of predicted top 15 cross-reactive peptides. Each row lists the peptide sequence and source gene. Color intensity indicates predicted binding strength (white to red).

area under the precision–recall curve (AUPRC) > 0.66; Fig. 3b and Supplementary Fig. 1a). This confirms that fine-tuned pLMs effectively capture the complex specificity rules encoded within the PRPs (Extended Data Fig. 4).

To assess whether incorporating α -chain information could improve performance, we trained paired $\alpha + \beta$ models and compared them with β -only counterparts across the full TCR panel. Inclusion of α -chain sequences did not significantly enhance predictive accuracy (median AUPRC, 0.85 versus 0.86; $P = 0.39$, Wilcoxon test; Extended Data Fig. 5a,b), indicating that α -derived features provided minimal additional predictive value in this β -dominated system. Consistent with these computational findings, α -chain swap activation assays showed that substituting α -chains (8.4a, 19.2a and 135.4a, all sharing AJ28) onto a common β -chain (19.2b) led to only minor quantitative differences in CD69 induction, with overall activation patterns closely tracking the β identity (Extended Data Fig. 5c,d). Together, these results confirm that, within this clonotypically constrained HLA-B*27:05-restricted family, conditional on a compatible α -chain, β -chain-encoded determinants are sufficient to capture the dominant features of peptide specificity, while α -chain variation primarily fine-tunes recognition rather than redefining it.

To probe the models' learned representations, gradient-based saliency methods identified peptide residues most influential for binding in TCRs 8.4, 19.2 and 4.3 (Fig. 3c). Similarly, the shuffling of amino acids at each position confirmed the critical hotspots necessary for accurate binding prediction (Supplementary Fig. 2). These *in silico* interaction hotspots faithfully recapitulated experimentally derived binding motifs and aligned with known structurally important contact residues (for example, P4, P5 and P8) across a range of different TCRs. This alignment with experimental data demonstrates that the pLMs accurately infer key determinants of TCR–pMHC interactions, including the influence of sequence position and amino acid variation on binding specificity.

To assess whether our models could generalize beyond the random synthetic peptide library and identify potential autoantigens in AS and AAU, we predicted the binding of >200,000 HLA-B*27:05-restricted 9-mer peptides from the human proteome (Fig. 3d and Methods). Our PRP-informed pLMs estimated peptide-binding probabilities for each TCR in the panel, identifying a subset of human peptides predicted to be highly cross-reactive, eliciting strong binding predictions for multiple TCRs. Ranking peptides by predicted cross-reactivity (Fig. 3e) revealed 15 candidates with considerable sequence diversity predicted to bind >33% of the TCRs. These findings indicate that our PRP-informed pLMs can identify potential immunodominant autoantigens and predict self-antigen cross-reactivity with the human proteome, warranting direct experimental validation of their capacity to induce T cell activation.

PRP-informed models predict T cell activation by human proteome peptides

While binding is a prerequisite, T cell activation is the physiologically critical readout for TCR–pMHC interactions. Therefore, we tested whether our pLM binding predictions could identify activating human peptides. Using the top 15 cross-reactive human peptides predicted by our models, we performed T cell activation assays (CD69 upregulation) with TCR-transduced SKW-3 cells (Fig. 4a, Supplementary Fig. 3 and Supplementary Table 1). Notably, despite being trained solely on peptide-binding data (the PRPs), our pLMs effectively discriminated between activating and nonactivating peptides (Fig. 4b,c and Supplementary Fig. 4). Our PRP-informed models significantly outperformed existing structural modeling metrics (that is, AlphaFold3 and tFold-TCR ipTM) in predicting activation (Fig. 4d and Supplementary Fig. 5; AUROC $P = 0.029$, AUPRC $P = 0.005$, Wilcoxon signed-rank test). We posit that pLMs (and likely other sequence-based methods) that leverage PRPs may implicitly capture context-dependent

patterns, including peptide conformational dynamics, which are often lost in static structural predictions, providing a more nuanced understanding of peptide binding and TCR engagement.

Across multiple TCRs, predicted binding scores correlated significantly with experimental T cell activation (Fig. 4c and Supplementary Fig. 6; Spearman $\rho > 0.5$, $P < 0.05$), including several novel candidate autoantigens. Notably, several peptides elicited stronger activation than a known HLA-B*27:05-restricted bacterial epitope (YEIH). A small subset of peptides, such as those derived from pregnancy-specific glycoprotein 5 (PSG5) and pre-mRNA processing factor 3 (PRPF3), activated the majority of tested TCRs (Fig. 4b), supporting the hypothesis of immunodominant, highly cross-reactive autoantigens in AS and AAU.

To further assess physiological relevance, we benchmarked our models against an independent set of human peptides selected solely on the basis of predicted HLA-B*27:05 binding. Specifically, we selected the top 20 strongest HLA-B*27:05-binding peptides predicted by NetMHCpan and experimentally tested their ability to activate the full panel of disease-related TCRs. None of these high-affinity self-peptides elicited measurable CD69 upregulation in any of the TCRs (Supplementary Fig. 7), confirming that strong MHC binding alone does not equate to TCR activation and highlighting the added value of PRP-informed models for capturing functionally relevant cross-reactivity.

Among the novel activating peptides, one derived from PSG5 was notable because of its anatomic expression. Single-cell RNA-seq data revealed that PSG5 is expressed in human iris pigment epithelial cells (Fig. 4e)²⁷. This expression pattern corresponds with the region of inflammation in AAU, the iris, which implicates PSG5 as a novel candidate autoantigen in AAU and potentially AS pathogenesis.

Direct *ex vivo* staining with PSG5–HLA-B27:05 tetramers demonstrated a statistically significant increase in PSG5-specific CD8⁺ T cell frequencies in individuals with HLA-B27⁺ axSpA/AAU compared to HLA-B*27⁺ healthy controls (Fig. 4f). As an internal control, SRY (HLA-B27-restricted flu peptide) tetramer staining revealed no significant difference in flu-specific CD8⁺ T cell frequencies between individuals with HLA-B27⁺ axSpA/AAU and controls, supporting the disease specificity of the PSG5 response. Furthermore, no sex-based differences were observed in PSG5-specific T cell frequencies, excluding the possibility that the response simply reflects physiological PSG5 exposure in female individuals (Extended Data Fig. 6)²⁶.

A crystal structure of the PSG5–HLA-B27:05–TCR 19.2 complex (Fig. 4g and Supplementary Fig. 8) revealed that PSG5 and YEIH peptides engage the TCR using highly similar binding patterns, with conserved docking geometry and contact residues. This structural similarity aligns with our machine learning models, which identified PSG5 as a highly cross-reactive peptide capable of activating multiple disease-associated TCRs.

Predictive generalization within a TCR neighborhood is enhanced by joint modeling

The identification of compelling candidate autoantigens such as PSG5, predicted by individual models trained on broad PRPs, and their recognition by multiple TCRs suggests that these TCRs may leverage shared or similar underlying principles for peptide interaction, which our models implicitly capture. To further understand the robustness of these predictions and delineate how our models might generalize TCR specificity, we next sought to systematically investigate the impact of TCR sequence variation on peptide recognition and model performance.

To dissect how TCR β -chain diversity influences knowledge transfer to new TCRs, we first constructed a 'TCR neighborhood' by engineering the 19.2 TCR with 1–3 amino acid substitutions in its CDR3 β region, while maintaining the original α -chain (Supplementary Fig. 9). Peptide-binding profiling revealed that these engineered TCRs exhibited highly similar binding specificities, confirming their status as

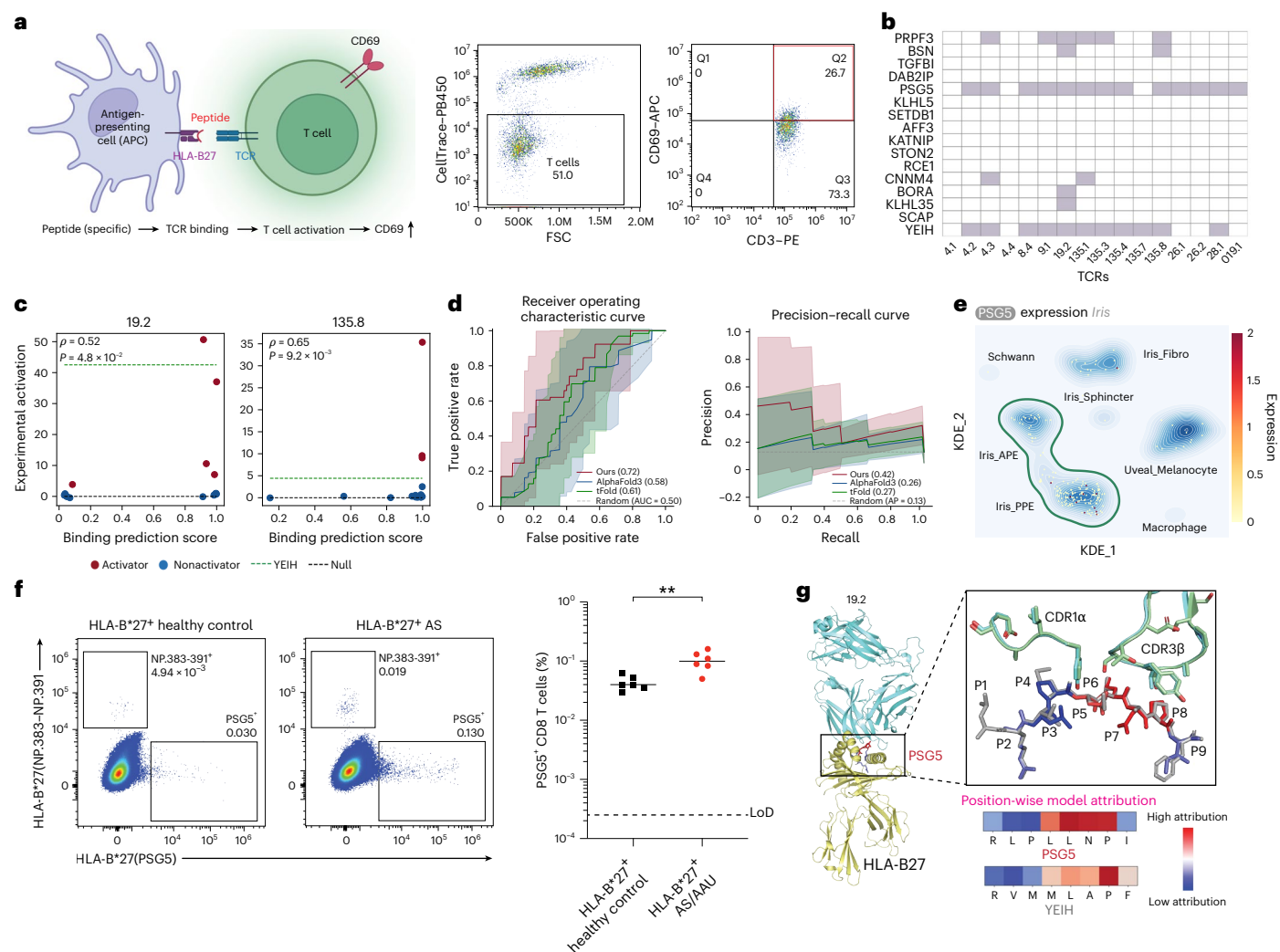


Fig. 4 | Experimental validation confirms model-predicted activating peptides. **a**, Schematic of peptide presentation assay. Peptides are presented by HLA-B27:O5 expressed K-562 cells to SKW-3 cells expressing defined TCRs. T cell activation is measured by CD69 upregulation by flow cytometry. Representative gating plots show activation readout. The image in **a** was created in BioRender; Wang, N. <https://biorender.com/oobm657> (2026). **b**, Summary of peptides experimentally validated as activators for each TCR. (binarization criteria in the Methods). **c**, Parity plots comparing predicted binding scores and experimental activation for TCRs 19.2 and 135.8. Each point represents a peptide (red, activators; blue, nonactivators). Dashed black line, null response; green dashed line, known activating peptide YeiH. Spearman ρ and two-sided P values are reported for each TCR. **d**, ROC and PR curves comparing performance of the peptide activation prediction model (red) against AlphaFold3 (blue) and tFold-TCR (green). Solid lines represent the mean performance and shaded

areas represent the mean $\pm$ s.d. **e**, Spatial transcriptomic mapping of PSG5 expression in human iris tissue, highlighting localized expression in iris pigment epithelial (Iris_PPE) cells. Data were reanalyzed from the cell atlas of the human ocular anterior segment²⁷. **f**, Representative flow cytometry of PBMCs from a healthy control (left) and HLA-B27 AS and/or AAU (right), stained with PSG5 and NP.383-391. Quantification of HLA-B27(PSG5)⁺ CD8⁺ T cells in peripheral blood from HLA-B27⁺ individuals acting as healthy controls ($n = 6$) and participants with HLA-B27⁺ AS and/or AAU ($n = 6$). Tetramer⁺ cells are shown as a percentage of all CD8⁺ T cells. **g**, Crystal structure of the PSG5-HLA-B27:O5-TCR (19.2) complex. Side (left) and close-up (right) views show peptide residues in the MHC groove and their TCR contact positions. Peptide residues are colored according to gradient-derived positional importance (blue, low; red, high). The color bar (bottom) indicates model-derived residue-wise attribution scores.

a functional neighborhood (Fig. 5a). Consequently, a pLM trained solely on wild-type (WT) 19.2 PRP data generalized effectively to the individual 19.2 mutants, with performance metrics for binding prediction differing by ~ 0.05 on average (AUPRC) when evaluated on their respective recognized peptides (Fig. 5b and Supplementary Fig. 1b). This generalizability indicates that these related TCRs indeed share core interaction principles.

However, despite this high accuracy in predicting peptide binding, the models trained on individual TCRs showed limited ability to predict T cell activation on our cross-reactive peptide panel (average AUROC = 0.64, average AUPRC = 0.53). This prompted us to test whether integrating data from the entire neighborhood could yield a more robust model. We hypothesized that jointly training a single

pLM on the combined PRPs of the 19.2 neighborhood would enhance predictive power for activation by capturing shared recognition principles (Fig. 5c). Notably, we found that a model jointly trained on all 19.2 variants demonstrated significantly improved performance (AUROC: $P = 0.03125$; and AUPRC: $P = 0.015625$) in predicting peptide-induced T cell activation for TCRs within this neighborhood, compared to models trained on individual TCRs (Fig. 5d and Supplementary Fig. 10). This joint model yielded refined binding prediction scores for the tested human peptides, indicating that integrating peptide recognition data from closely related receptors enhances the identification of activating ligands by improving the resolution of fine-specificity determinants or providing a more robust consensus binding profile for the TCR neighborhood.

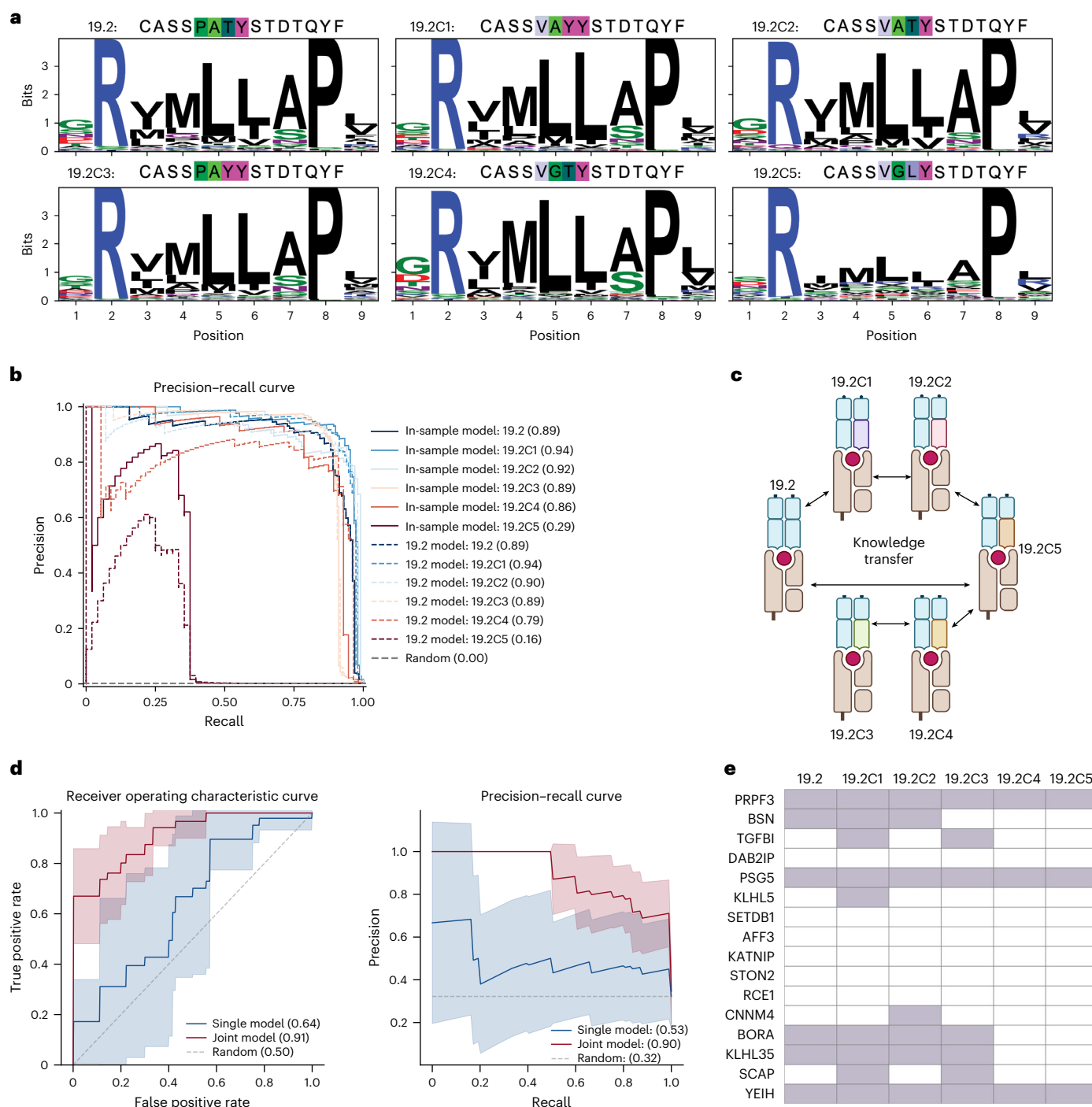


Fig. 5 | TCR models generalize to sequence variants and benefit from neighborhood-based joint training. a, CDR3 β sequences and corresponding peptide-binding motifs for WT 19.2 and its five mutants, visualized as sequence logos. **b**, PR curves for predicting peptide on the deep sequencing test set of each mutant TCR. Solid lines represent models trained individually on each variant (in sample), while dashed lines show predictions using the model trained on the parental TCR 19.2. **c**, Illustration of knowledge transfer among related TCRs in a sequence neighborhood. Schematic of mutant CDR3 β design based on TCR 19.2, with individual variants (19.2C1–19.2C5) containing amino acid

substitutions in the β -chain. The image in **c** was created in BioRender; Yeh, H. <https://biorender.com/55g4tqg> (2026). **d**, ROC and PR curves comparing the predictive performance on activation data for each TCR variant using models trained on individual TCR data (solid lines) versus a single model jointly trained on the entire AS19.2 neighborhood (dashed lines), demonstrating the benefit of joint training. Shaded areas represent the mean $\pm$ s.d. **e**, Summary of peptides experimentally validated as activators for TCR 19.2 and its mutants (binarization criteria in the Methods).

Notably, several immunodominant human peptides previously identified (for example, derived from PRPF3, PSG5 and the YEIH epitope) consistently activated these 19.2 CDR3 β mutants, despite the engineered perturbations (Fig. 5e and Supplementary Fig. 11). This persistent recognition across related TCR variants further supports

their potential pathological significance in AS. While these experiments illuminate generalization within a local TCR space defined by sequence similarity, they also underscore the need for metrics such as functional distance to predict performance when encountering entirely new, potentially divergent TCRs.

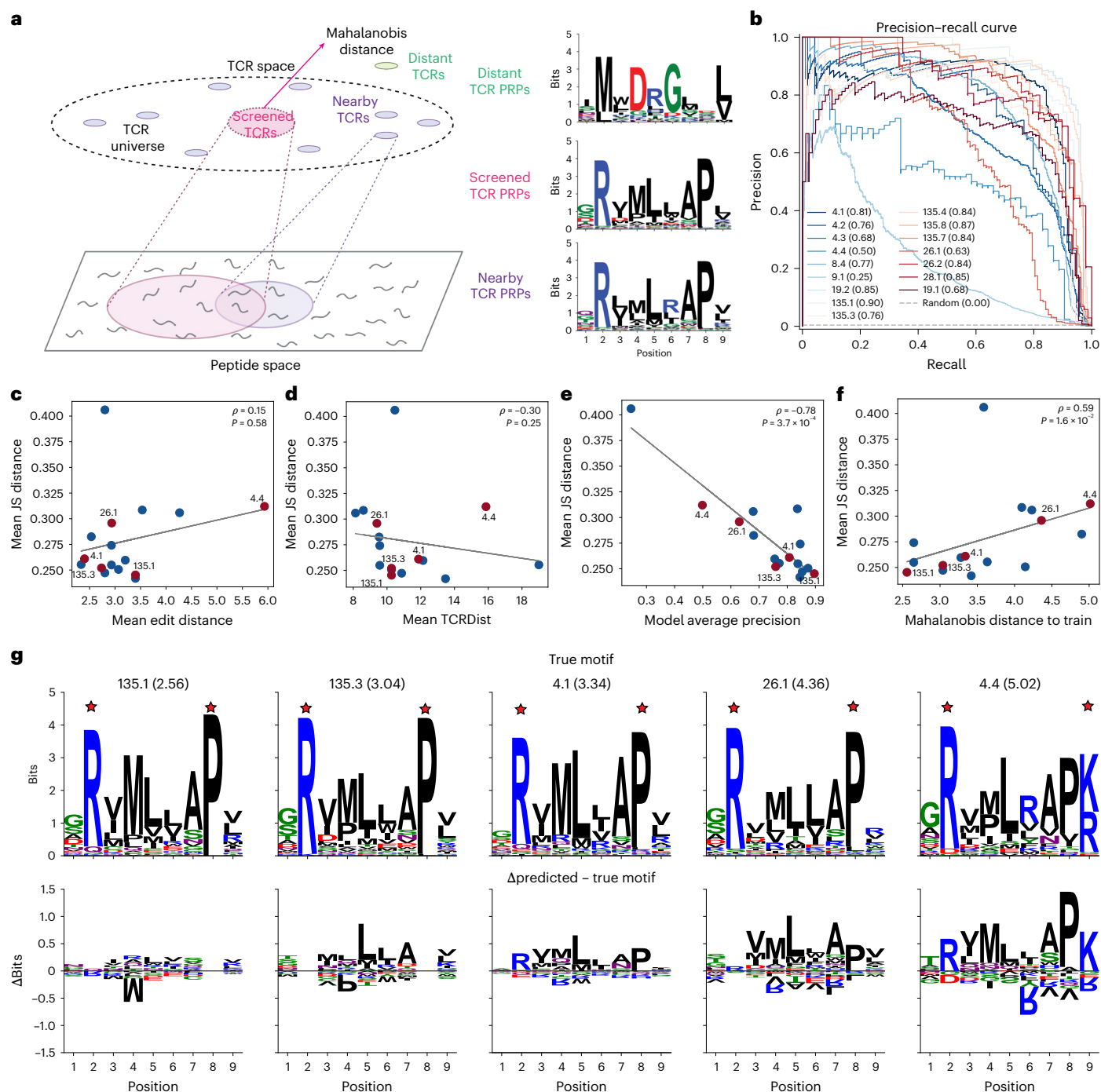


Fig. 6 | Predicting model generalization to novel TCRs using peptide recognition fingerprints and embedding distance.

a, Conceptual schematic of TCR and peptide recognition space. Screened TCRs occupy a defined region of this space, while new TCRs may lie nearby or further away. Functional distance is defined by divergence in predicted PRPs. Mahalanobis distance provides a model-intrinsic estimate of how far a new TCR lies from the training distribution ('out-of-distribution' status). Representative PRPs are shown for screened and nearby TCRs from this study and for a functionally distant TCR based on previously published peptide recognition data²⁶. The image in **a** was created in BioRender; Wang, N. <https://biorender.com/xsxyek1> (2026). **b**, LOO TCR cross-validation performance across all TCRs, assessed by PR curves for

peptide-binding prediction. **c–f**, Correlation between PRP divergence (mean JS distance) and various metrics: edit distance (**c**), TCRdist (**d**), model AP in held-out prediction (**e**) and Mahalanobis distance to train (**f**). For all panels, Spearman ρ and two-sided P values are reported. In **f**, Mahalanobis distance in the model's learned embedding space correlates with experimental PRP divergence, providing a quantitative 'confidence score' for predictions on new TCRs. TCRs shown in **g** (135.1, 135.3, 4.1, 26.1 and 4.4) are highlighted in red in **c–f**. **g**, Motif prediction performance for five representative held-out TCRs. Top, true peptide-binding motifs. Red stars indicate peptide positions fixed during library construction. Bottom, Δ bits (predicted – true) at each peptide position. Mahalanobis distance values used in **f** are labeled after each TCR name.

Generalizing across TCRs: performance limits and the utility of model-intrinsic uncertainty

Building on these findings, we next addressed the fundamental challenge of generalizing predictions to entirely novel TCRs not

encountered during training. We hypothesized that the 'functional distance' between a new TCR and the existing training data, quantified by the divergence of their PRPs, would be the key determinant of predictive success, rather than simple sequence similarity (Fig. 6a)²⁸.

To test this, we used a rigorous leave-one-out (LOO) TCR cross-validation strategy across our full TCR panel (training on $n - 1$ TCRs, testing on the held-out TCR; Fig. 6b, Supplementary Fig. 1c and Methods).

We found that standard TCR sequence similarity metrics (for example, CDR3 edit distance and TCRDist) failed to predict functional divergence (mean JS distance of PRPs to training set) (Fig. 6c,d). Conversely, AUPRC strongly anticorrelated with the actual functional distance (PRP divergence) between the held-out TCR and the training set (Spearman $\rho \approx -0.78$, $P < 0.001$; Fig. 6e). This establishes that relatedness in peptide recognition patterns, not sequence similarity, is the primary determinant of predictive transferability to new TCRs.

We then asked whether the model itself could provide an intrinsic estimate of its reliability for unseen TCRs. To this end, we used the Mahalanobis distance as a multivariate measure of how far a new TCR lies from the cloud of training TCRs in the learned joint TCR–peptide embedding space. This metric, which we interpret as a model-estimated degree of out-of-distribution behavior, correlated significantly with experimentally measured functional distance (PRP divergence) (Spearman $\rho \approx 0.59$, $P \approx 0.02$; Fig. 6f). Thus, unlike sequence-based heuristics, our model provides a quantitative, peptide-recognition-aware measure of its own reliability for novel TCRs. This intrinsic uncertainty offers a principled framework to assess prediction reliability and strategically guide experimental profiling of peptide recognition toward functionally distinct TCRs where model confidence is low.

The utility of this metric is demonstrated by the model's motif predictions (Fig. 6g and Extended Data Fig. 7). For held-out TCRs with high Mahalanobis distance, predicted motifs deviate noticeably from the true motifs, whereas, for TCRs with low Mahalanobis distance, the model generates motifs nearly identical to the true motif. In other words, when the embedding places a new TCR close to familiar training receptors, motif and activation predictions are more accurate. This demonstrates that our model's confidence score is directly linked to its capacity for biologically meaningful generalization, making it a reliable tool to guide future TCR profiling.

Discussion

This work presents an integrated experimental–computational platform for profiling and predicting TCR recognition, centered on empirically determined PRPs. By combining high-throughput peptide–MHC yeast display with fine-tuned pLMs, we link sequence to function at high resolution. The resulting PRP data resource and predictive models provide a functionally grounded approach for mapping TCR specificity, discovering disease-associated antigens and defining the principles that govern predictive generalization within a constrained TCR family.

A cornerstone of our findings is the high-resolution, empirical demonstration that TCRs can be organized into clusters based on their PRPs. While convergent evolution of TCR specificity is a known phenomenon, our platform provides quantitative detail, revealing distinct peptide recognition neighborhoods even among TCRs restricted by a single MHC allele (HLA-B*27:05). Our framework shares conceptual parallels with prior models describing how TCR motifs vary systematically across epitope space²⁹, reinforcing the broader machine learning intuition that TCR–peptide relationships can be productively viewed through mapping of recognition landscapes. In parallel, our paired $\alpha + \beta$ modeling and α -chain swap experiments show that, within this TCR family, CDR3 α primarily fine-tunes recognition and modulates response magnitude rather than redefining core peptide specificity, which is dominated by the β -chain. However, these findings should not be interpreted to mean that α -chains are dispensable for TCR specificity more broadly. Our β -chain-based predictions hold only conditional on the presence of a structurally compatible α -chain that permits productive pMHC engagement. Together, these observations provide a framework for extending PRP-based profiling to more diverse α – β combinations in future work. The resulting PRP-defined classification offers a biologically meaningful complement to sequence-based

grouping methods such as GLIPH and TCRdist and highlights the value of applying advanced machine learning to dense peptide-binding data to capture the nuanced grammar of TCR–pMHC interactions.

Importantly, this framework extends beyond binding to predict functional T cell activation by peptides from the human proteome, even though the models were trained on binding-enriched yeast display data rather than direct functional measurements. We then validated predictions against T cell activation readouts, thereby beginning to bridge the gap between binding and function. In this setting, PRP-informed pLMs demonstrated strong performance in discriminating activating from nonactivating peptides, notably surpassing metrics from existing structural modeling approaches (for example, AlphaFold3 and tFold-TCR's ipTM) in this specific predictive task. This finding does not diminish the fundamental importance of structure in molecular recognition. Rather, it suggests that our pLMs, by learning from vast peptide-binding datasets, implicitly capture complex features, potentially reflecting subtle energetic contributions, conformational dynamics or kinetic parameters not fully encapsulated by static structural compatibility metrics alone, which are essential determinants of cellular activation thresholds. The successful identification of novel candidate autoantigens for AS, most notably PSG5 expressed in iris tissue, highlights the immediate translational power of this approach for uncovering disease-associated ligands. This specific finding provides a compelling, testable hypothesis for a molecular link between HLA-B*27 and a key extraarticular manifestation of AS.

Our study also clarifies how predictions generalize to novel TCRs within this disease-associated restricted system. We find that predictive accuracy is better explained by functional distance (PRP divergence) to the training set rather than by sequence similarity alone. This shifts the emphasis, in this context, from sequence proximity to functional relatedness when assessing where models are likely to perform well. To this end, we introduce a model-intrinsic uncertainty metric, the Mahalanobis distance within the learned TCR–peptide embedding space, as a quantitative, peptide-recognition-aware estimate of prediction reliability for novel TCRs. This metric substantially outperformed sequence similarity in forecasting generalization difficulty (correlation with PRP divergence). It provides a principled strategy to gauge confidence in predictions for unseen receptors, thereby enabling the efficient prioritization of experimental efforts toward TCRs with distinct peptide recognition patterns and informative regions of the vast TCR universe. The ability to further refine predictions through joint modeling within peptide-recognition-defined TCR neighborhoods, as shown with the 19.2 variants, suggests an iterative path toward resolving even fine-specificity differences, guided by this understanding of relatedness in peptide binding.

Our results should not be interpreted as contradicting sequence-based molecular mimicry at the peptide level, which has been convincingly demonstrated in prior work²⁴. Instead, they extend this framework by showing that even highly similar TCRs can diverge in their PRPs, underscoring that mimicry and cross-reactivity cannot always be inferred from TCR sequence similarity alone, particularly within dense, disease-associated clonotype families.

We also recognize important limitations of our current implementation. This study focuses on a highly related family of HLA-B*27:05-restricted, disease-associated TCRs. While repertoire-scale sequence-based clustering methods such as GLIPH or TCRdist excel at global repertoire organization, they are inherently limited in resolution within dense, high-similarity clonotype families. In this context, PRPs provide fine-grained functional distinctions that sequence alone cannot resolve. Additionally, our β -centric modeling applies specifically to this clonotypically constrained family in which α – β pairing has already been subject to strong biological selection; generalization to diverse α – β pairings will require expanded profiling. From a technical perspective, the yeast display peptide libraries used here incorporated fixed anchor residues (arginine at P2 and proline at P8) to balance

diversity with biological relevance and relied on standard stabilizing substitutions in HLA-B27:05 (Y84A, L5M, H114Y, A153D and C67S) to enhance folding and surface display. These design choices improve selection efficiency and data quality but restrict the accessible peptide space and may subtly influence peptide binding and orientation. All candidate antigens identified in this setting were, therefore, validated in the context of WT HLA-B27:05 using independent biochemical, structural and cellular assays.

Despite these constraints, the integrated platform presented here represents a generalizable engine for discovery. The principles of defining specificity through deep PRPs and leveraging these data with advanced machine learning can be applied to diverse MHC alleles, to the study of immune repertoires in cancer and infectious disease and to other immune receptor systems. The ability of our models to predict activating peptides while providing an intrinsic confidence score can guide the rational design of TCR-based immunotherapies, helping to engineer receptors with tailored specificities and enhanced safety profiles.

In conclusion, we establish a peptide-recognition-centric framework for defining and predicting TCR specificity. This integrated platform and its associated data resources are poised to accelerate our fundamental understanding of adaptive immunity and serve as a resource for the broader immunology and biotechnology communities.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-026-03128-x>.

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Methods

Selection of TCRs

The 16 disease-derived TCRs analyzed in this study were previously obtained and characterized from AS and AAU samples using previously reported approaches. For clarity, they fall into three categories, outlined below.

Motif-guided TRBV9⁺ enrichment (public family). For TCR 4.1, T cells from the peripheral blood of an individual with HLA-B*27⁺ AS were expanded in vitro for 28 days using anti-CD3 and IL-2 followed by flow sorting on TRBV9⁺ CD8 T cells for single-cell TCR sequencing (scTCRseq)²⁴. For TCRs 8.4 and 9.1, TRBV9⁺ CD8 T cells from the synovial fluid of individuals with HLA-B*27⁺ AS were flow-sorted and subjected to scTCRseq²⁴. These belong to the well-described BV9-Y/FSTDQ-BJ2.3/ TRAV21 public family.

Non-BV9 example (broadened motif). For TCR 019.1, immune cells from the aqueous humor of a participant with HLA-B*27⁺ AAU were subjected to scTCRseq²⁴. This clonotype does not carry the BV9-Y/FSTDQ-BJ2.3 motif but instead represents an alternative β -chain background paired with TRAV21, broadening the family beyond BV9 usage.

YeiH tetramer-guided isolation. For TCRs 19.2, 26.1, 26.2, 28.1, 135.1, 135.3, 135.4, 135.7 and 135.8, T cells from the peripheral blood of HLA-B*27⁺ AAU participants were expanded in vitro for 8 days using YeiH-pulsed autologous peripheral blood mononuclear cells followed by flow sorting on YeiH tetramer⁺ CD8 T cells for scTCRseq²⁶. These include both TRBV9 and TRBV5 clonotypes (TRBV5-1 and TRBV5-5, IMGT nomenclature), ensuring diversity beyond the canonical BV9 family.

Engineered TCR variants and α -chain swaps

To assess model generalization within a local neighborhood, we generated five engineered variants of TCR 19.2 by introducing one to three amino acid substitutions into its CDR3 β loop while maintaining the original α -chain. These engineered receptors (19.2C1–19.2C5) were expressed and profiled in parallel, providing a controlled neighborhood of closely related clonotypes for assessing model generalization.

To validate the β -centric modeling approach, we generated chimeric TCR constructs by swapping AJ28-matched α -chains. The α -chain sequences of TCRs 8.4 and 135.4 were synthesized and cloned into a pHRLentiviral vector containing the WT 19.2 β -chain. These chimeric receptors were expressed in SKW-3 cells (DSMZ, ACC53) and assayed for activation alongside the parental WT TCRs.

Flow cytometry using PSG5–HLA-B27:05 tetramers to stain CD8 T cells from AAU, AS and healthy controls

Study approval. All human participants were enrolled in accordance with Declaration of Helsinki principles and using protocols approved by the institutional review board of Washington University in St. Louis (Institutional Review Board 201912043 and 202212047). Written informed consent was received from participants before inclusion in the study.

Participants. Participants with HLA-B27⁺ AAU fulfilled the Standardization of Uveitis Nomenclature Working Group classification criteria for HLA-B27⁺ AAU³⁰. Participants with HLA-B27⁺ AS satisfied the modified New York criteria³¹. Two individuals with AAU also met diagnosis of nonradiographic axial spondyloarthritis by Assessment of SpondyloArthritis International Society (ASAS) classification criteria for axial spondyloarthritis (all participant demographics provided in Supplementary Table 2)³². Healthy controls were recruited through the Volunteers for Health registry at Washington University in St. Louis. All were free of autoimmune diseases. HLA-B*27 positivity was confirmed through PCR in the laboratory.

Biospecimen collection. Peripheral blood samples were obtained by venipuncture and collected into EDTA tubes. Peripheral blood mononuclear cells (PBMCs) were isolated by Ficoll-Hypaque density gradient centrifugation. PBMCs were cryopreserved in FBS containing 10% dimethyl sulfoxide (DMSO) and stored at -140°C until analysis²⁶.

Flow cytometry. Frozen PBMCs were thawed and washed twice with RPMI + 10% FBS. A total of 10 million cells were used for flow cytometric assessment of HLA-B*27-restricted CD8⁺ T cells specific to PSG5 and NP.383-391 (SRYWAIRTR). PSG5 tetramers were conjugated using streptavidin–PE (Agilent, PJRS25-1) and used at a 1:50 dilution. NP.383-391 tetramers were conjugated using streptavidin–BV650 (BioLegend, 405232) and used at a 1:100 dilution²⁴. Cells were washed with PBS, stained with fixable viability dye (Thermo Fisher Scientific) at 4°C for 20 min and then washed once with FACS buffer (PBS containing 1% FBS). Cells were resuspended in Fc block (Thermo Fisher Scientific) for 10 min at room temperature. They were then incubated with tetramers and surface antibodies (all antibody reagents provided in Supplementary Table 3) for 30 min at room temperature. Cells were washed twice with FACS buffer and fixed with 2% PFA for flow cytometry. Data were collected on Cytex Aurora flow cytometer (Cytex Biosciences) and analyzed with FlowJo (version 10; Tree Star).

Expression of AS/AAU TCR in Expi293 mammalian expression system

TCR α -chains and β -chains were cloned into the pD649 mammalian expression vector. The α -chain construct included a C-terminal basic leucine zipper, while the β -chain construct contained a C-terminal acidic leucine zipper and a BirA biotinylation tag. Both chains were cotransfected into Expi293 cells (Thermo Fisher, A14527) following the manufacturer's protocol³³.

After a 4-day expression period, cultures were centrifuged at 2,000g for 10 min to collect the supernatant. The supernatant was mixed with an equal volume of PBS (pH 7.2, Gibco) and incubated with 2 ml of Ni-NTA agarose slurry (Qiagen) per 50 ml of supernatant at 4°C overnight with gentle rotation. The mixture was passed through a gravity column to isolate resin-bound protein. The resin was washed with 30 column volumes (CVs) of PBS containing 20 mM imidazole and eluted with 10 CVs of PBS containing 200 mM imidazole.

The eluted protein was concentrated using a Millipore filter (30-kDa molecular weight cutoff) to a final volume of $\sim 500\ \mu\text{l}$ and subsequently biotinylated overnight using BirA. Final purification was performed by size-exclusion chromatography using a Superdex 200 Increase column (GE Healthcare) equilibrated in PBS pH 7.2.

Fractions were analyzed for heterodimeric TCR formation and successful biotinylation by SDS–PAGE (under reducing and nonreducing conditions) and streptavidin shift assays. Peak fractions were pooled, quantified by absorbance at 280 nm, aliquoted and stored at -80°C until use.

Construction and selection of yeast-displayed peptide–HLA-B*27:05 libraries

The peptide–HLA-B27:05 library was displayed on yeast as a single-chain trimer (SCT) comprising a 9-mer peptide, β 2-microglobulin (β 2m) and the HLA-B27:05 heavy chain, linked by flexible linkers. To enhance stability, the construct included previously reported substitutions (Y84A, L5M, H114Y and A153D with C67S to prevent aberrant disulfide formation) and was assembled following established protocols²⁴.

Peptide libraries were generated using primers encoding pre-defined codons (all primers are provided in Supplementary Table 4). To ensure optimal binding to HLA-B*27:05 and specificity toward AS-associated TCRs, anchor residues at positions P2 and P8 were fixed as arginine and proline, respectively. The remaining positions were encoded using NNK degenerate codons to introduce diversity. Each 9-mer library was constructed independently and fused to a C-terminal

MYC epitope tag for flow cytometry detection. Library complexity was estimated by colony counting after yeast electroporation.

Transformed yeast was grown in SDCAA medium (pH 4.5), induced in SGCAA medium (pH 4.5) and subjected to iterative selection. Before positive selection, negative depletion was performed by incubating yeast ($10\times$ library diversity) with streptavidin-coated magnetic beads at 4 °C for 1 h to remove nonspecific binders. Cells were passed through LS columns (Miltenyi) under a magnetic field and washed three times; the flowthrough was collected for subsequent enrichment.

For positive selection, yeast were incubated with 400 nM biotinylated soluble TCR preconjugated to streptavidin-coated MACS beads at 4 °C for 3 h. Bound cells were retained on LS columns, eluted and expanded overnight in SDCAA medium. This induction and selection cycle was repeated for a total of four rounds. Selected yeast populations were stored at 4 °C and used for next-generation sequencing analysis.

In addition, TCRs 4.2, 4.3 and 4.4 were analyzed using previously published peptide–HLA-B*27:05 library data, in which P2 and P9 were fixed while P8 was unconstrained²⁴.

Deep sequencing of pHLA libraries

Yeast DNA was extracted from approximately 1×10^7 yeast cells per selection round using the ZymoPrep II Miniprep Kit (Zymo Research). Peptide-encoding regions were PCR-amplified over 30 cycles using primers containing Illumina sequencing adaptors, unique 8-nt unique molecular identifiers and round-specific barcodes to distinguish between selection rounds (Supplementary Table 4).

PCR products were purified by agarose gel extraction and quantified using the Qubit dsDNA HS assay kit (Thermo Fisher Scientific). Libraries were sequenced on an Illumina MiSeq platform using a 2 $\times$ 150-bp v2 kit. Forward reads were demultiplexed in Geneious Prime 2024.0.7 on the basis of barcode sequences to separate individual selection rounds. Reads containing frameshifts or premature stop codons were removed from further analysis.

Identical peptide sequences were counted and clustered using custom Python scripts and Unix shell commands. For each TCR, peptide counts from selection rounds 1–4 were aggregated into a structured dataset, which served as input for downstream machine learning-based prediction and analysis.

Deep sequencing binding binarization and dataset preparation

Peptide counts from selection rounds 1–4 were used to binarize binding labels. Peptides were designated as positive binders if they had a count ≥ 5 in round 4. Negative binders were defined as peptides with a count of zero across rounds 2–4.

T cell activation in coculture assays

TCRs were cloned into the pHR lentiviral vector. Full-length α -chain and β -chain constructs were cloned separately and cotransfected with packaging plasmids pMD2.G and pSPAX2 into HEK293T cells (American Type Culture Collection (ATCC), CRL-3216) using Fugene (Promega) in DMEM supplemented with 10% FBS. Lentiviral supernatants were isolated 60 h after transfection, clarified by centrifugation and used to transduce 1.5×10^6 CD8⁺ SKW-3 cells by spinfection in the presence of 8 $\mu\text{g ml}^{-1}$ polybrene (Millipore Sigma). Transduced cells were cultured in RPMI complete medium and expanded for 5–7 days. TCR surface expression was validated by flow cytometry using PE anti-CD3 (UCHT1, BioLegend, 300408) or PE anti-TCR (IP26, BioLegend, 984702) antibodies.

Similarly, full-length HLA-B*27:05 and $\beta 2\text{m}$ were cloned into the pHR vector and packaged in HEK293T cells under identical conditions. Lentivirus was collected 60 h after transfection and used to transduce 1.5×10^6 K-562 cells (ATCC, CCL-243) following the same spinfection protocol. After expansion for 5–7 days, surface HLA expression was assessed by staining with APC anti-HLA-A/B/C antibody (W6/32, BioLegend, 311410) and analyzed by flow cytometry.

For antigen presentation, HLA-B*27:05-transduced K-562 cells were labeled with CellTrace violet (Invitrogen) for 20 min and incubated with 15 predicted cross-reactive peptides at 37 °C in 5% CO₂ for 1–2 h. Peptides ($\geq 70\%$ purity, Elim Biopharm) were dissolved in DMSO at 10 mM and 1 μl of peptide stock was added to 100 μl of K-562 cells to reach a final concentration of 100 μM . After peptide loading, cells were washed with RPMI and cocultured with TCR-transduced SKW-3 cells at a 1:1 ratio for 14–18 h.

T cell activation was assessed by staining for APC anti-CD69 (FN50, BioLegend, 310910) and PE anti-CD3 (OKT3 or UCHT1, BioLegend, 317308 or 300408), followed by flow cytometric analysis (CytoFLEX, Beckman Coulter) and analysis using GraphPad Prism (versions 8.4 and 9.0).

T cell activation data binarization

To normalize T cell activation data, the CD69% of CD3⁺ cells stimulated with a null control peptide was subtracted from the CD69% of CD3⁺ cells stimulated with each test peptide. This normalization was performed independently for each replicate, followed by averaging across replicates. For binarization, peptides eliciting $>3\%$ CD69 expression were classified as activators.

Expression of HLA-B*27 as inclusion bodies in *Escherichia coli*

The HLA-B*27:05 heavy chain and human $\beta 2\text{m}$ were expressed in *E. coli* as inclusion bodies. The respective genes were cloned into the pET26b vector and transformed into *E. coli* BL21(DE3) competent cells. A single colony was picked and cultured overnight in 10 ml of LB medium supplemented with 50 $\mu\text{g ml}^{-1}$ kanamycin at 37 °C with shaking at 250 rpm for 12–16 h. The overnight culture was then inoculated into 1 L of LB medium containing 50 $\mu\text{g ml}^{-1}$ kanamycin and incubated at 37 °C, 300 rpm until the optical density at 600 nm (OD₆₀₀) reached 0.5–0.7.

Protein expression was induced by adding IPTG to a final concentration of 1 mM, followed by continued shaking at 37 °C for an additional 3 h. Bacterial cultures were harvested by centrifugation at 6,000g for 15 min at 4 °C. Pelleted cells were resuspended in 50 ml of lysis buffer (50 mM Tris-HCl pH 8.0, 100 mM NaCl, 10 mM DTT, 1% Triton X-100, 5 mM MgCl₂ and 0.2 mM PMSF) and lysed by sonication using a 2 min on–off cycle, repeated three times.

The lysate was centrifuged at 10,000g for 15 min at 4 °C. Subsequently, the pellet was resuspended in 50 ml of wash buffer (50 mM Tris-HCl pH 8.0, 100 mM NaCl, 0.5% Triton X-100, 1 mM EDTA, 1 mM DTT and 0.2 mM PMSF) and subjected to the same sonication cycle (2 min on, 2 min off, repeated four times), followed by centrifugation at 10,000g for 15 min. This wash step was repeated once more with wash buffer.

The inclusion body pellet was solubilized in 25 ml of solubilization buffer containing 8 M urea, 20 mM Tris-HCl pH 8.0, 0.5 mM EDTA and 1 mM DTT. Insoluble debris was removed by centrifugation at 60,000g for 30 min at 4 °C and the clarified supernatant was collected. Solubilized protein was aliquoted and stored at –80 °C until use in refolding experiments.

Refolding of pMHC

The refolding buffer was prepared with 100 mM Tris-HCl pH 8.0, 400 mM arginine, 0.5 mM oxidized glutathione, 5 mM reduced glutathione and 2 mM EDTA. Peptide (20 mg) was dissolved in DMSO and added to 1 L of refolding buffer.

For each liter of refolding buffer, 20 mg of HLA-B*27:05 heavy chain inclusion body and 20 mg of human $\beta 2\text{m}$ inclusion body were premixed and added dropwise into the refolding buffer under gentle stirring. The refolding mixture was transferred into dialysis tubing (Spectrum Labs) and dialyzed against 10 L of 20 mM Tris-HCl pH 8.0, with buffer changes every 12 h for a total of four exchanges.

Following dialysis, refolded pMHC complexes were filtered through a 0.22- μm membrane, concentrated to ~1 ml and subjected

to ion-exchange chromatography using a Mono Q column (GE Healthcare). Final quality assessment was performed by size-exclusion chromatography on a Superdex 200 Increase column (GE Healthcare) using an ÄKTA purifier system.

Crystallization of AS TCR–pHLA-B*27:05 complexes

AS TCRs expressed in GnTI[−] Expi293 cells (Thermo Fisher, A39240) were treated with Endo H, 3C protease and carboxypeptidases A and B at 4 °C for 24 h to remove *N*-linked glycans and C-terminal leucine zipper tags. The reaction mixture of TCR and refolded pMHC was subsequently purified by size-exclusion chromatography using a Superdex 200 column (GE Healthcare). Protein complexes were crystallized by sitting-drop vapor diffusion: AS19.2–YEIH–HLA-B*27:05 was at 10 mg ml^{−1} using 0.2 M sodium malonate pH 6.4, 17% PEG 3350 and AS19.2–PSG5–HLA-B*27:05 at 17 mg ml^{−1} with 0.2 M sodium formate and 20% PEG 3350. Crystals were cryoprotected with addition of 30% glycerol and flash-cooled in liquid nitrogen for data collection.

Structure determination and refinement

Diffraction data were collected at Stanford Synchrotron Radiation Lightsources (SSRL) beamline 12-2. Intensities were processed using XDS³⁴. The YEIH structure was determined by molecular replacement using the PHASER (version 2.5.5) program with components of PDB 7N2R as the search models and the PSG5 structure used the rebuilt TCR 19.2 chains from the YEIH structure with HLA chains from PDB 7N2R. Model building was performed manually in Coot (version 0.9.6), and structure refinement was conducted using the PHENIX (version 2.5.5) software suite^{35,36}. Simulated annealing omit electron density maps are presented in Supplementary Fig. 8. Structural analyses were carried out with programs from the CCP4 package (version 7.0)³⁷. Crystallographic software was installed and maintained by SBGrid³⁸. Data collection and refinement statistics are provided in Supplementary Table 5. Final coordinates and structure factors were deposited to the RCSB Protein Data Bank under accession codes 9PBG and 9PBH. Diffraction images were deposited to the SBGrid data bank under accession numbers 1178 for YEIH and 1179 for PSG5.

Motif analysis and sequence logos

Positive binder peptides for each TCR were used to generate motif using MoDec (version 1.2)³⁹ where the best motif was selected across five runs allowing for a single motif with the background amino acid frequencies drawn from the human proteome.

JS distances were calculated using Scipy (`scipy.spatial.distance.jensenshannon`) by taking the sum of the distances for all putative TCR contact residues (positions 0, 2, 3, 5 and 6). MDS was then performed using sklearn (`sklearn.manifold.MDS`) with `n_components = 2` using the computed JS distances.

Machine learning architecture

Our model architecture is centered on a single pretrained encoder ESM2 (650-million-parameter model), to encode and extract representations for both the CDR3 α/β and antigen peptide sequence to obtain sequence-wise representations. Each sequence was independently embedded using mean-pooled position-wise representations. Additionally, the position-wise embeddings concatenated with one-hot encoded sequence was further processed and passed through a six-layer transformer encoder. The output of the transformer encoder and mean-pooled position-wise representations were concatenated to form the final embeddings for each peptide and CDR3 α/β . For our β -chain models, a joint representation was then computed as the outer product between the CDR3 and peptide for the sequence-wise embeddings and position-wise representations (two channels). For our α/β -chain models, the outer product was computed for each CDR3 and peptide and concatenated in the channel dimension (four channels). Joint embeddings were then processed through a three-layer

convolutional neural network, followed by a single-layer multilayer perceptron to predict binding probability. Binary cross-entropy loss was used for optimization. All model parameters, including the ESM2 encoder, were fine-tuned during training.

Pretraining on VDJdb

Before training on deep sequencing data, we pretrained the model on VDJdb. We extracted CDR3 α/β -peptide pairs belonging to MHC class I. We used the cognate pairs provided as positives. Negative pairs were generated by uniformly sampling noncognate peptides from the VDJdb pool each epoch. An equal number of positive and negative examples were sampled each epoch. The data were split into training (95%) and validation (5%) sets. The model was trained using the Adam optimizer (learning rate: 1×10^{-6}) until convergence, with early stopping based on a validation patience of ten epochs.

Training single-TCR models

Fine-tuning was performed on each binarized TCR dataset using the pretrained VDJdb model. Each dataset was split into training (70%), validation (15%) and test (15%) sets. Training used the same hyperparameters as during pretraining. To address class imbalance, where negative samples greatly outnumber positives, we applied a weighted binary cross-entropy loss, with class weights set inversely proportional to the number of positive and negative examples in the training set. This ensured that each class contributed equally. Performance was evaluated on the held-out test set using the average precision (AP) and AUROC. For TCR 19.2, we additionally evaluated the model on its five mutant variants (C1–C5) to assess generalization to mutant CDR3 β variants.

Training joint and LOO TCR models

We fine-tuned the VDJdb-pretrained model on a concatenated dataset comprising all 16 individual TCR datasets. The same 70:15:15 train-validation-test split was used to enable fair comparison to individual TCR models. For the joint 19.2 model (Fig. 5d), we combined the WT and five mutant (C1–C5) datasets in training. To assess generalizability, we trained 16 LOO models, each using data from 15 TCRs and evaluating on the held-out TCR. As TCRs 28.1 and 135.7 share the same CDR3 β , both were excluded whenever either was held out to prevent data leakage. Consequently, although 16 LOO models were trained, only 15 were unique. Training hyperparameters were identical to those used in prior experiments.

Generating human proteome 9-mer peptide library and nominating peptides for activation study

We used NetMHCpan-4.1 and the UniProt human reference proteome (UP000005640) to predict all 9-mer peptides with HLA-B*27:05-binding potential. Peptides with percentage rank < 2 (weak binders) and percentage rank < 0.5 (strong binders) were subselected, yielding 293,467 unique peptides. On the basis of prior knowledge that the tested TCRs exhibit a strong preference for proline at P8 of the peptide, we further filtered the dataset to include only peptides containing proline at P8, yielding 12,357 unique peptides. To nominate peptides for functional testing, we applied each single-TCR model to score this peptide library. For each TCR, peptides with predicted binding scores > 0.95 were selected. We then subselected peptides predicted to bind at least five TCRs. Ranking each of the subselected peptides by the number of TCRs predicted to bind, the top 15 peptides were selected. The maximum number of TCRs that a single peptide was predicted to bind was 12. For model performance evaluation and benchmarking against other models (AUROC and AP), we excluded TCRs 4.1, 4.4 and 135.7, as none of the selected peptides activated these TCRs. To benchmark against peptide to MHC affinity-based prediction, we selected the top 20 peptides from the human proteome with the highest predicted affinity for HLA-B*27:05 using NetMHCpan-4.1. These peptides were

experimentally tested for T cell activation to serve as high-affinity nonactivating controls.

Calculating TCR distances

To assess relatedness between TCRs and TCR neighborhoods in the LOO setting, we computed several distance metrics. The JS distance was computed position-wise between amino acid distributions derived from peptide enrichment profiles obtained through yeast-display screening and averaged across non-excluded peptide positions. For each included position i , amino acid distributions p_i and q_i , representing the peptide-binding motifs of two TCRs being compared, were compared using the following definitions:

$$d_{JS}(p, q) = \frac{1}{|I|} \sum_{i \in I} \sqrt{\frac{D_{KL}(p_i \| m_i) + D_{KL}(q_i \| m_i)}{2}}, \quad m_i = \frac{1}{2}(p_i + q_i),$$

$$D_{KL}(p_i \| m_i) = \sum_{a \in \mathcal{A}} p_i(a) \log \frac{p_i(a)}{m_i(a)}, \quad I = \{1, \dots, L\} \setminus \{2, 8, 9\}$$

Here, D_{KL} denotes the Kullback-Leibler divergence, $\mathcal{A}$ denotes the amino acid alphabet and I denotes the set of included peptide positions. Positions P2, P8 and P9 were excluded from distance calculations because P2 corresponds to a canonical MHC anchor position, while P8 and P9 were fixed in subsets of peptide libraries during library construction, resulting in limited amino acid variability at those positions among positive binders. This motif-based JS distance was used as the ground-truth measure of similarity between TCRs. Edit distance was computed using CDR3 β amino acid sequences only. TCRdist3 was computed using TCR β -chain information only, including the CDR3 β sequence together with the corresponding V- and J-gene annotations, to ensure a fair comparison with our TCR β -centric model. Mahalanobis distance was calculated as

$$d_{Mah}(x, \mu) = \sqrt{(x - \mu)^T S^{-1} (x - \mu)}$$

where x denotes the sequence-level representations of the held-out CDR3 β sequence after the ESM module, μ denotes the mean vector of training-set sequence-level representations after the ESM module and S^{-1} denotes the Moore-Penrose pseudoinverse covariance matrix computed from the training-set sequence-level representations.

Because TCRs 28.1 and 135.7 share identical CDR3 β sequences, each TCR was excluded from the other's distance calculations to maintain consistency with the LOO training procedure.

AlphaFold3 and tFold-TCR predictions

To benchmark against existing complex structure-prediction algorithms, we evaluated both AlphaFold3 (a general multimolecular structure predictor) and tFold-TCR (an algorithm specifically trained for TCR-pMHC complex modeling). For each peptide in our activation study, we supplied the full sequences for each component of the TCR-pMHC complex as input to each method using default inference settings. Both AlphaFold3 and tFold generate interface confidence metrics intended to reflect the reliability of predicted intermolecular contacts. We extracted the ipTM score from each model and used this score as the structure-based predictor of TCR activation.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The atomic coordinates and structure factors for the TCR-pMHC complexes were deposited to the RCSB Protein Data Bank under accession codes PDB [9PBG](#) (AS19.2-YEIH-HLA-B27:05) and PDB [9PBH](#) (AS19.2-PSG5-HLA-B27:05). Diffraction images were deposited to the SGBrd data bank with accession codes [1178](#) for YEIH and [1179](#) for PSG5.

The processed PRPs, including peptide sequences and enrichment counts for each TCR and the naive library, were deposited to the Dryad Digital Repository (<https://doi.org/10.5061/dryad.h70rxwdzk>)⁴⁰. All other data supporting the findings of this study, including source data for main and supplementary figures, T cell activation assay results and participant demographics, are available within the article and its Supplementary Information.

Code availability

The custom code used for fine-tuning the pLMs, predicting TCR-peptide interactions and performing the analyses presented in this study is publicly available. The complete codebase, including data and scripts for model training and evaluation, is available on GitHub (<https://github.com/akds/PRP-TCR-Specificity>) and archived on Zenodo (<https://doi.org/10.5281/zenodo.19190730>)⁴¹.

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Acknowledgements

We thank T. Doukov at SSRL for assistance with data collection. Use of the SSRL, SLAC National Accelerator Laboratory, is supported by the US Department of Energy (DOE), Office of Science, Office of Basic Energy Sciences under contract no. DE-AC02-76SF00515. The SSRL Structural Molecular Biology Program is supported by the DOE Office of Biological and Environmental Research and by the National Institutes of Health (NIH), National Institute of General Medical Sciences (P30GM133894). H.Y. is supported by the Medical

Scientist Training Program Training Grant T32GM150375. J.U. is supported by NIH award T32-AR007279. M.A.P. is supported by NIH award K08-AR079593. A.A.K. is supported by NIH award DP2AI177884, a Breakthrough T1D/LRA/NMSS Common Mechanisms of Autoimmunity Insight Award (1-SRA-2025-1755-A-N) and a Chan Zuckerberg Investigator Award. K.C.G. is an investigator with the Howard Hughes Medical Institute and is supported by NIH R01AI103867 and the Parker Institute for Cancer Immunotherapy (PICI). This work was delivered as part of the MATCHMAKERS team, supported by the Cancer Grand Challenges partnership financed by Cancer Research UK (CGCATF-2023/100006) and the National Cancer Institute (OT2CA297242).

Author contributions

A.A.K. and K.C.G. supervised the research. N.W., H.Y., B.L., J.P., X.Y., A.A.K. and K.C.G. designed the experiments, performed the statistical analyses and visualized the data. N.W., X.C., C.W. and L.D.L. conducted the yeast display experiments, deep sequencing data analysis, target validation, TCR-pMHC complex purification and T cell activation. H.Y., B.L., J.P. and A.A.K. developed the algorithm for peptide prediction. K.M.J., N.W. and X.X. conducted the structural studies. I.R., J.U. and M.A.P. conducted the processing of samples from individuals with AS and AAU, single-cell T cell sequencing analysis and TCR tetramer staining of SCTs. A.A.K. wrote the initial draft of the manuscript.

N.W. and H.Y. contributed to writing, revisions and figure preparation. All authors edited and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

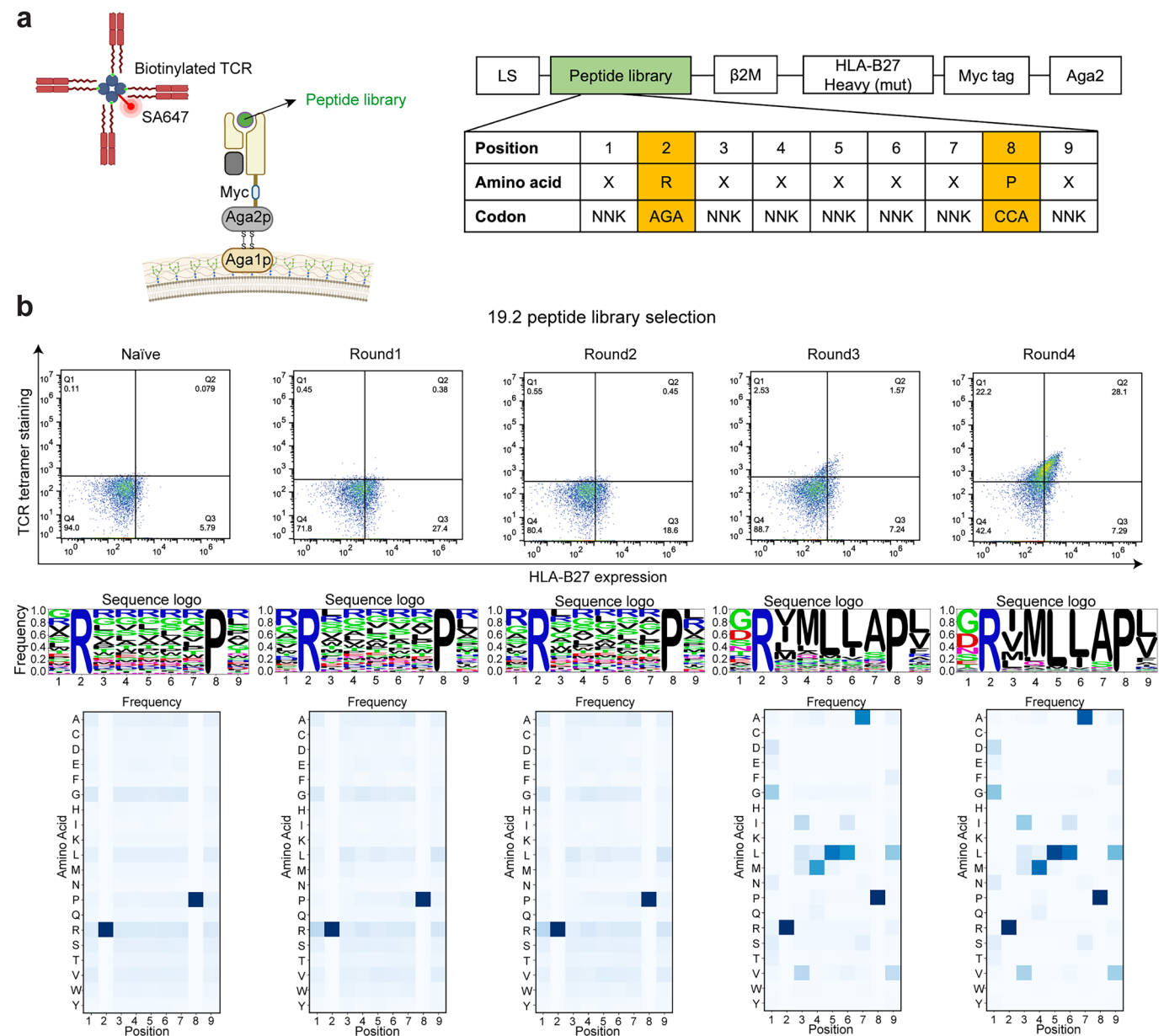
Extended data is available for this paper at <https://doi.org/10.1038/s41587-026-03128-x>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41587-026-03128-x>.

Correspondence and requests for materials should be addressed to Aly A. Khan or K. Christopher Garcia.

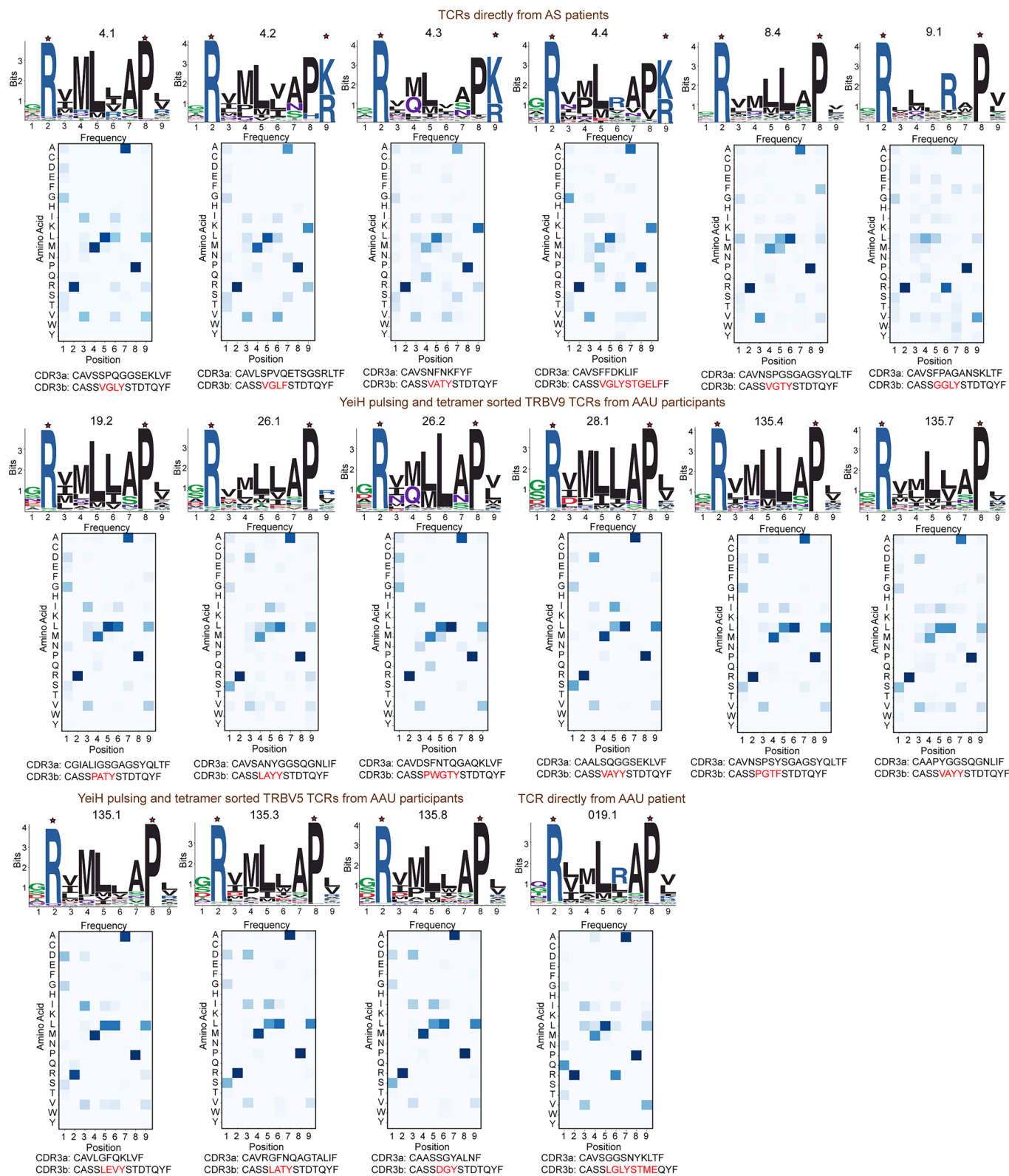
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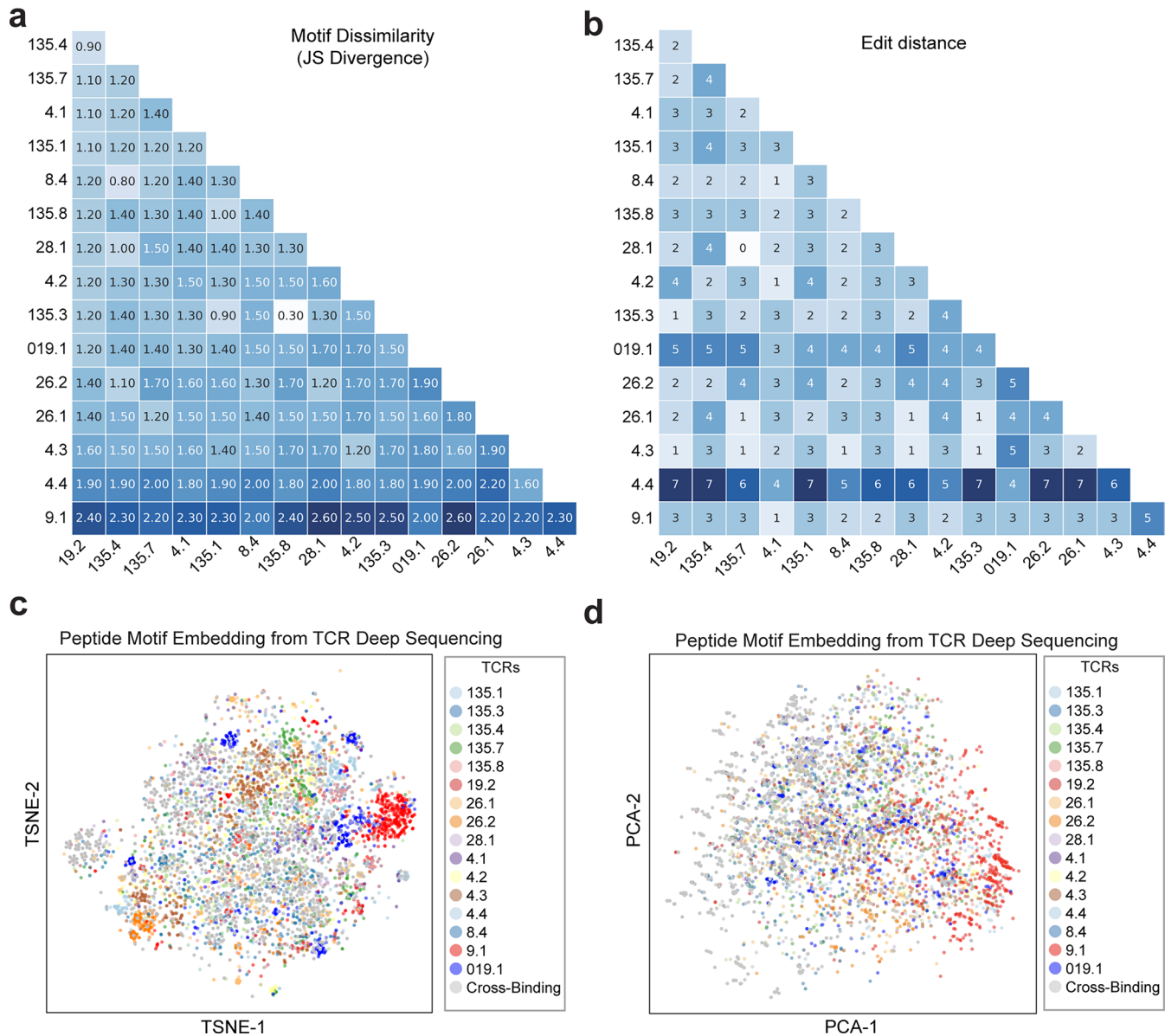


Extended Data Fig. 1 | Peptide library selection and enrichment over rounds of yeast display. a, Schematic of yeast display selection using biotinylated TCR and streptavidin-647 to isolate binding peptides from a library of 9-mers with fixed P2 arginine and P8 proline residues. Created in BioRender. Wang, N. (2026) <https://BioRender.com/cxabe2h>. **b,** Representative flow

cytometry plots showing enrichment of binding peptides across four rounds of selection for TCR19.2. Corresponding sequence logos and position-wise amino acid enrichment heatmaps highlight the evolution of peptide motifs over selection rounds.

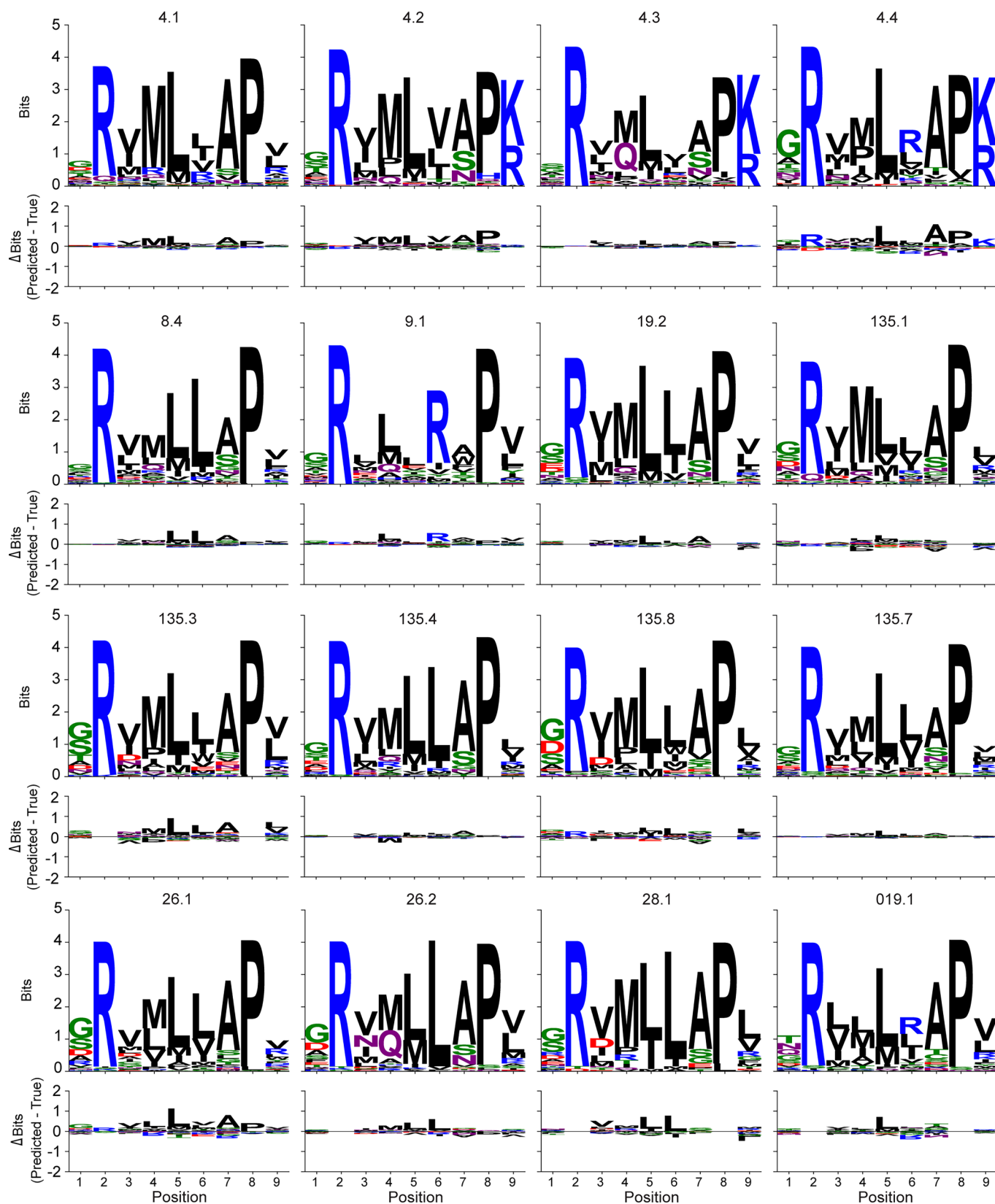


positions fixed during library construction. For TCRs 4.2, 4.3, and 4.4, peptide libraries were constructed with fixed residues at position 2 (Arg) and position 9 (Lys/Arg), whereas for all other TCRs, libraries were fixed at position 2 (Arg) and position 8 (Pro). CDR3α and CDR3β sequences are indicated for each TCR.

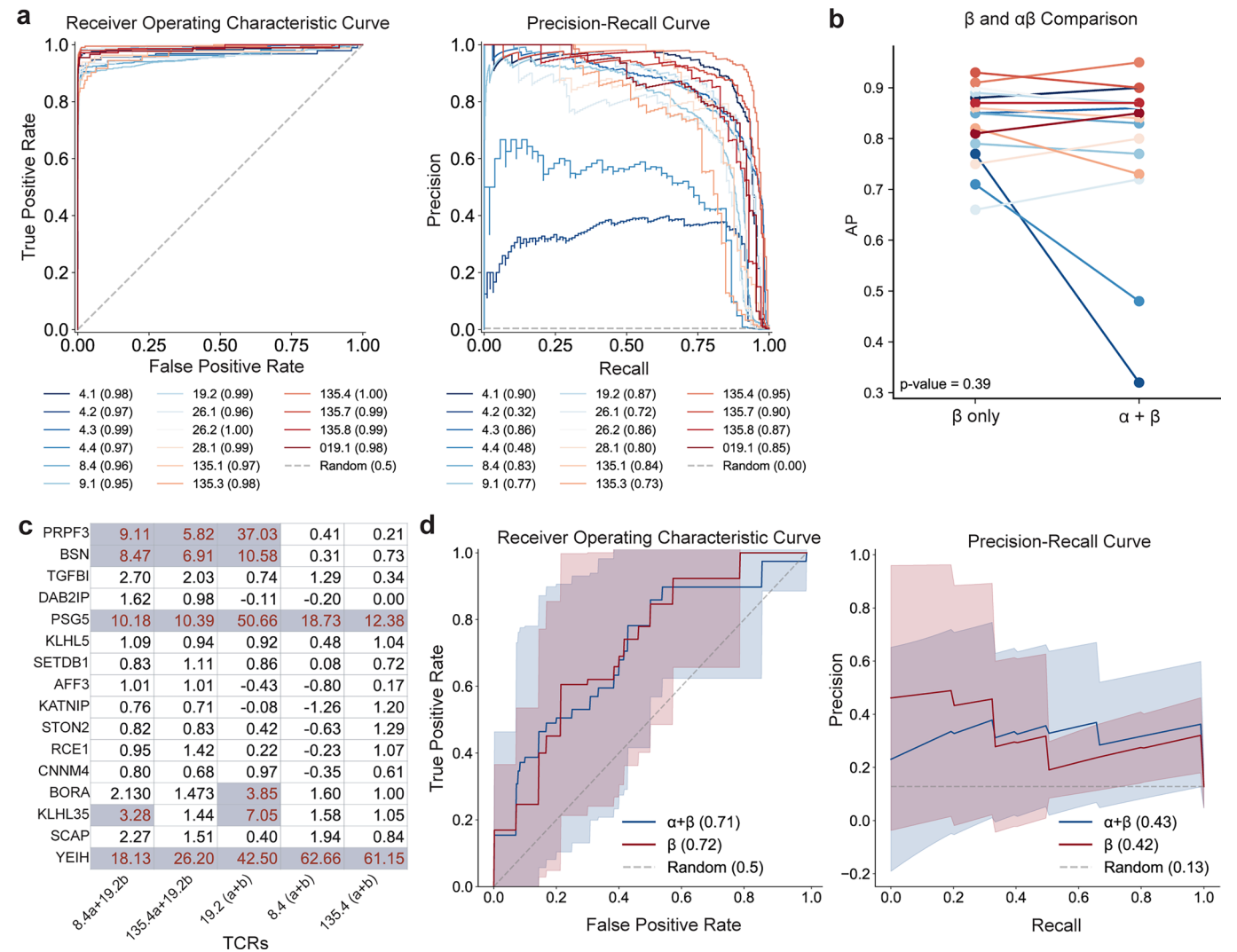


Extended Data Fig. 3 | Pairwise comparison of TCR similarity. a, Jensen–Shannon (JS) divergence between peptide binding motifs. **b,** Edit distance based on CDR3 β amino acid sequences. Darker hues indicate greater dissimilarity. **c,d,** t-SNE and PCA projections of enriched peptides from all TCRs based on

their sequence embeddings. Each point represents a peptide, colored according to the TCR(s) it binds to. Clustering of peptides from different TCRs reflects shared recognition patterns.

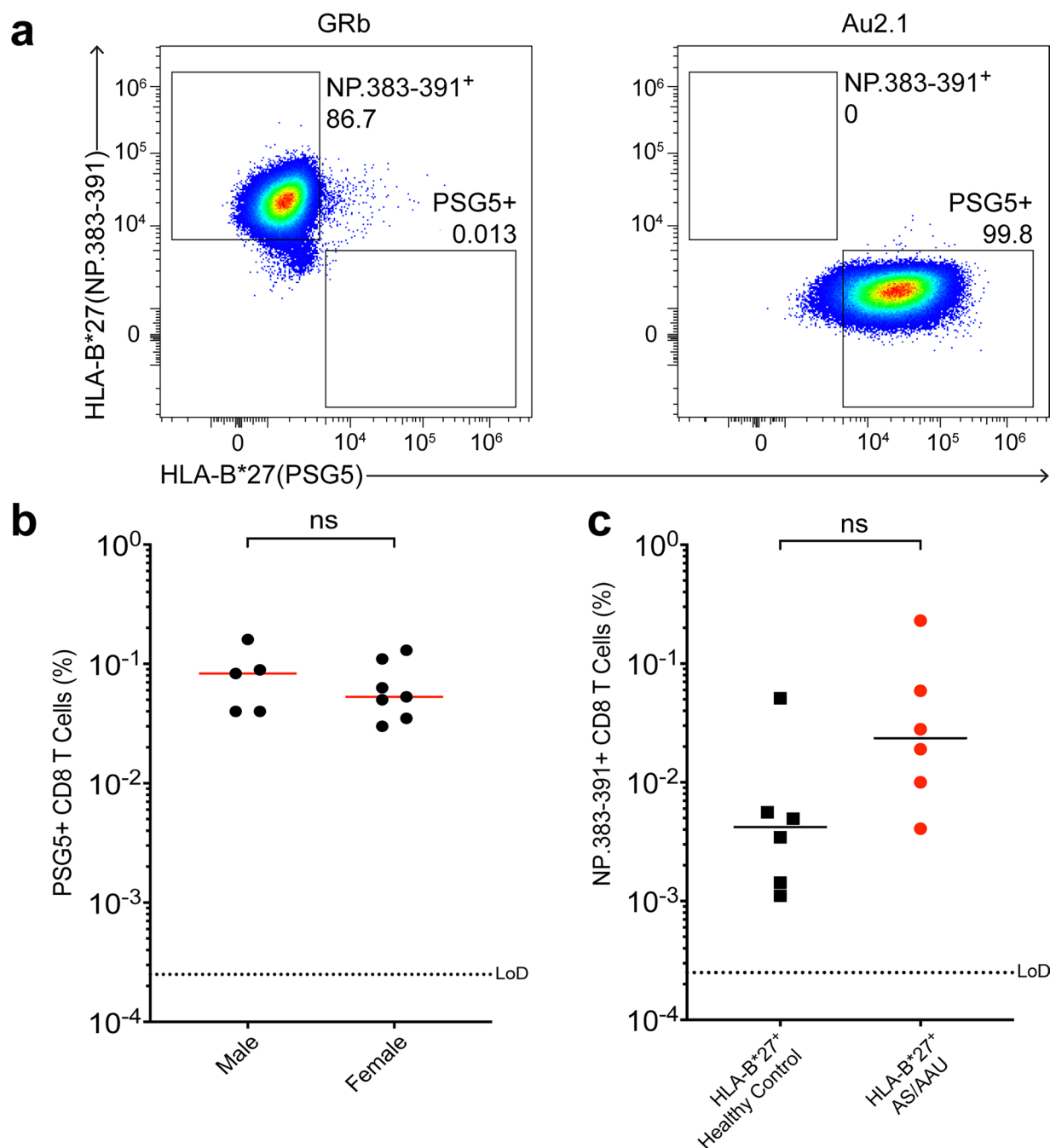


Extended Data Fig. 4 | Predicted peptide-binding motifs from single-TCR models. The top row shows the peptide-binding motifs predicted by individual TCR-specific models (see Methods). The bottom row displays the difference between the predicted and experimentally derived (true) motifs for each TCR.



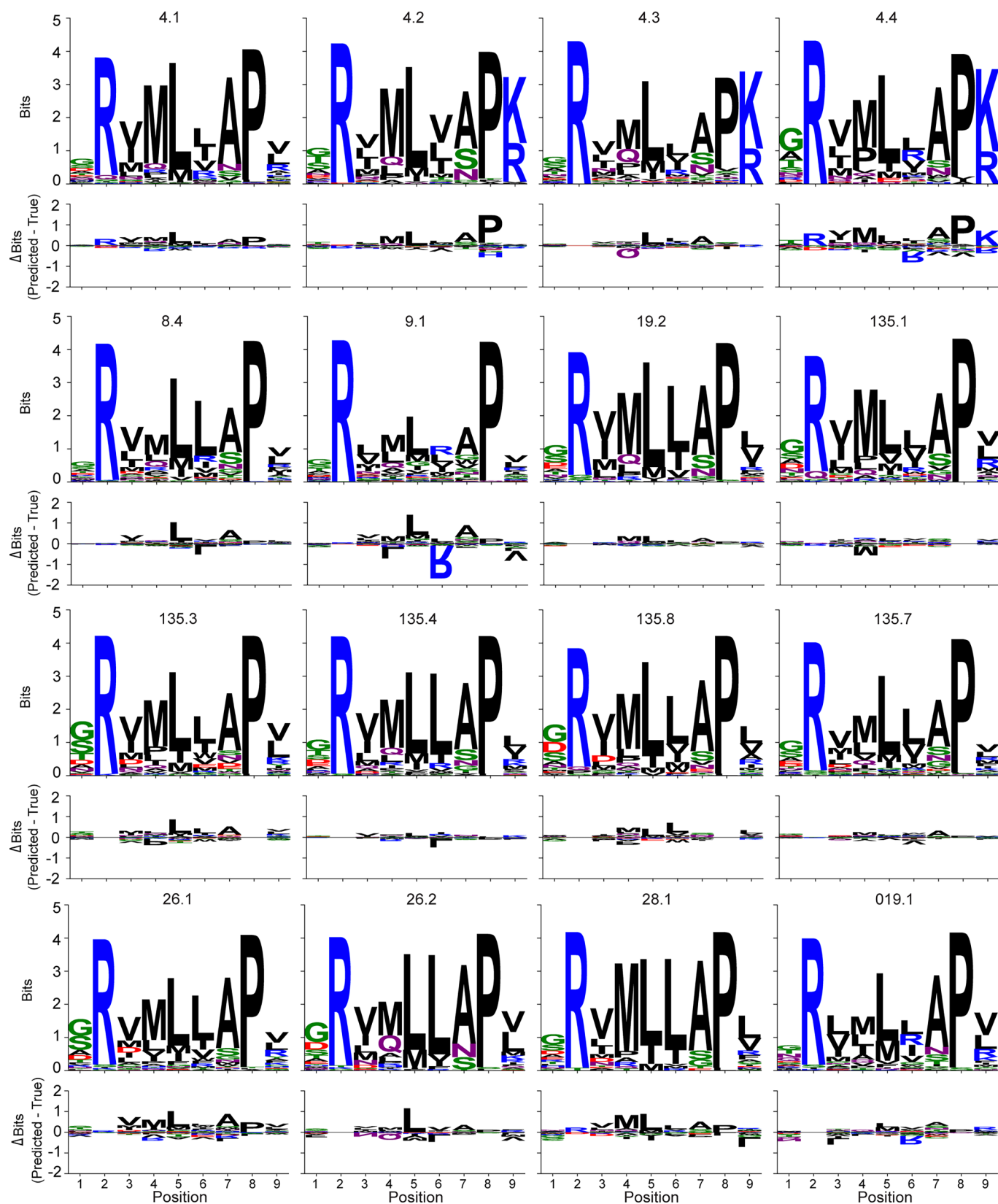
Extended Data Fig. 5 | Evaluation of α -chain contribution to peptide recognition and model performance. **a**, Receiver operating characteristic (ROC) and precision-recall (PR) curves for $\alpha + \beta$ models across the TCR panel. **b**, Comparison of PR AUC values between β -only and $\alpha + \beta$ models ($n = 12$ individual TCRs; two-sided Wilcoxon test, $p = 0.39$). Statistics were derived by comparing the two model architectures on the same held-out test sets for each TCR. **c**, α -chain swap activation assays using a fixed β -chain (19.2b) with

substituted α -chains (8.4a, 19.2a, and 135.4a; AJ28). Heatmap values indicate mean CD69 activation across three independent experiments, with peptides inducing $>3\%$ CD69 expression classified as positive. **d**, Receiver operating characteristic (ROC) and precision-recall (PR) curves comparing performance of the peptide activation prediction model between β -only (red) and $\alpha + \beta$ (blue) models. Solid lines represent the mean performance, and shaded areas represent the mean $\pm$ s.d.



Extended Data Fig. 6 | PSG5-specific CD8⁺ T cell detection using tetramer staining and comparison by sex. **a**, Validation of PSG5 and NP.383-391 tetramers for identifying pathogenic CD8⁺ T cells. The AAU-derived expanded clonotype Au2.1 demonstrates specific binding to HLA-B27(PSG5). The influenza-specific TCR GRb binds HLA-B27(NP383.391) but not PSG5, serving as comparative control. **b**, Frequencies of HLA-B27(PSG5)⁺ CD8⁺ T cells in peripheral blood from

HLA-B27⁺ healthy controls and participants with AS and/or AAU, stratified by sex (n=5 males, n=7 females). **c**, Frequencies of HLA-B27(NP.383-391)⁺ CD8⁺ T cells in peripheral blood from HLA-B27⁺ individuals acting as healthy controls (n=6) and participants with HLA-B27⁺ AS and/or AAU (n=6). Statistical significance was assessed using the two-sided Mann-Whitney test (**b**, **c**).



Extended Data Fig. 7 | Predicted motifs from leave-one-out (LOO) single-TCR models. The top row shows peptide-binding motifs predicted by LOO-trained TCR models (see Methods). The bottom row displays the difference between the predicted motifs and the experimentally derived (true) motifs for each TCR.

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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
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☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
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Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

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Data collection	Accuri (BD Biosciences); Cytoflex (Beckman Coulter); Miseq (illumina); SSRL Beamline BL12
Data analysis	Geneious Prime 2024.0.7; FlowJo v10; GraphPad Prism 8.4 and 9.0; XDS; PHASER v2.5.5; COOT v0.9.6; PHENIX v1.12; CCP4 package v7.0; MoDec v1.2; Structural visualization and refinement tools maintained via SGrid; Custom code for deep sequencing data processing and TCR specificity prediction available at: https://github.com/akds/PRP-TCR-Specificity

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

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Processed peptide recognition profiles (PRPs), T cell activation results, and patient metadata are available in the article and Supplementary Information. Crystal structures were deposited in the Protein Data Bank under accession codes 9PBG and 9PBH. Diffraction images are available at SGrid (DOIs: 10.15785/

SBGRID/1178, 10.15785/SBGRID/1179). Protein sequences for target prediction were from UniProt UP000005640. The processed Next-Generation Sequencing (NGS) data, including peptide counts for all TCR selections and naive libraries, have been deposited in the Dryad Digital Repository (<https://doi.org/10.5061/dryad.h70rxwdzk>). Custom code for deep sequencing data processing and TCR specificity prediction available (<https://github.com/akds/PRP-TCR-Specificity>) and archived on Zenodo (<https://doi.org/10.5281/zenodo.19190730>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Sex information was collected and is summarized in Supplemental Table 2.

Sex

HC252 Male

HC289 Male

HC310 Female

HC327 Female

HC343 Female

HC770 Female

AI028 Female

AI041 Male

AI065 Female

AI126 Female

AI206 Male

AI246 Male

Sex-based comparisons were performed for T cell clonal frequencies as shown in Extended Data Fig. 6b. No significant sex-based differences were observed in model performance.

Reporting on race, ethnicity, or other socially relevant groupings

Race or ethnicity information was not collected for this study. Samples were anonymized and obtained solely based on HLA genotype and disease status relevant to study goals.

Population characteristics

Participants included individuals diagnosed with HLA-B27+ acute anterior uveitis (AAU) or HLA-B27+ ankylosing spondylitis (AS), based on the Standardization of Uveitis Nomenclature (SUN) criteria and modified New York criteria, respectively. Two AAU participants also met the ASAS classification criteria for non-radiographic axial spondyloarthritis (nr-axSpA). Healthy controls were HLA-B*27+ and free of autoimmune diseases. All participant demographics are detailed in Supplemental Table 2.

Recruitment

Patients were recruited through clinical studies at Washington University in St. Louis. Healthy controls were recruited via the Volunteers for Health registry at the same institution. HLA-B*27 positivity was confirmed by PCR in the laboratory.

Ethics oversight

All human participants were enrolled in accordance with Declaration of Helsinki principles and using protocols approved by the institutional review board of Washington University in St. Louis (IRB 201912043 and 202212047). Written informed consent was received from participants prior to inclusion in the study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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Sample size

No formal sample size calculation was performed. Sample sizes were determined based on standard practices in the field and reflect practical limitations of human PBMC availability. For in vitro T cell activation assays, representative clones were used to validate model predictions. Sample size is consistent with prior studies using similar methods for validation.

Data exclusions

No data were excluded from the analysis unless they failed quality control thresholds (e.g. low sequencing depth or abnormal fluorescence background). All exclusion criteria were pre-established and applied consistently across samples.

Replication

Key findings were replicated across independent TCR clones and patient-derived PBMCs. Computational predictions were validated by yeast display, tetramer staining, and functional T cell activation assays in multiple donors. Figures include representative and/or repeated experiments, as indicated in legends.

Randomization

This was not an interventional study. No treatment groups or placebo controls were involved. For computational analyses, data were

Randomization randomly split into training and test sets to assess model performance.

Blinding Investigators were blinded to the clinical group allocation during data collection and initial processing. All patient and healthy control samples were processed using anonymized alphanumeric identifiers. The association between sample IDs and clinical status was only unblinded during the final statistical integration and figure preparation.

Reporting for specific materials, systems and methods

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Materials & experimental systems

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☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Antibodies used for flow cytometry immunophenotyping, T cell sorting are listed in Supplementary Table 3, including target, fluorochrome conjugate, clone name, vendor, and catalog number. Specifically, antibodies against CD3 (OKT3, ThermoFisher, 363-0037-42), CD4 (RPA-T4, ThermoFisher, 11-0049-42), CD8 (SK1, Cytex, R7-20238), CD19 (HIB19, ThermoFisher, 11-0199-42), TCR $\gamma\delta$ (B1.1, ThermoFisher, 11-9959-42), TCR V α 7.2 (3C10, Biolegend, 351703), and TCR V α 24-J α 18 (6B11, Biolegend, 342905) were used. In addition, the following reagents were used for functional assays: APC anti-human CD69 Antibody (FN50, Biolegend, 310910), PE anti-human CD3 Antibody (OKT3, Biolegend, 317308), PE anti-human CD3 Antibody (UCHT1, Biolegend, 300408), APC anti-HLA-A, B, C antibody (W6/32, BioLegend, 311410), PE anti-TCR antibody (IP26, BioLegend, 984702)
Validation	Antibodies listed in Supplementary Table 3 were used for immunophenotyping and FACS of patient-derived PBMC CD8+ T cells. The additional antibodies and proliferation dye listed above were used in T cell activation assays to assess surface marker expression. They are all commercial and validated by the vendors.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	Expi293 (ThermoFisher, A14527), GNTI-expi (ThermoFisher, A39240), 293T (ATCC, CRL-3216), SKW-3 (DSMZ, ACC53) and K-562 cells (ATCC, CCL-243).
Authentication	Cell lines were periodically authenticated (at least once per year) using short tandem repeat analysis.
Mycoplasma contamination	Cell lines were periodically tested for Mycoplasma contamination (at least once per year) using mycoplasma detection kit (Biotool).
Commonly misidentified lines (See ICLAC register)	None of the cell lines used are listed in the ICLAC database.

Plants

Seed stocks	No related to plants
Novel plant genotypes	No related to plants
Authentication	No related to plants

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
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Methodology

Sample preparation

Peripheral blood mononuclear cells (PBMCs) were isolated from patient blood samples using density gradient centrifugation. Cells were stained with fluorochrome-conjugated antibodies for surface marker analysis or sorted for CD8+ T cells using FACS. For yeast selections, yeast cells were washed with MACS buffer before staining and flow cytometry.

Instrument

Flow cytometry data were collected using a Cytex Aurora spectral flow cytometer and FACS sorting was performed on a BD FACSaria II cell sorter.

Software

Flow cytometry data were analyzed using FlowJo software 10.

Cell population abundance

Sorted CD8+ T cell populations typically displayed >90% purity post-sort, as determined by flow cytometric re-analysis. For yeast staining, at least 100,000 cells were analyzed. T cell assays typically used 10,000 cells per population.

Gating strategy

Standard gating strategy was applied including FSC/SSC to exclude debris, singlet gating, live/dead exclusion, and gating on CD3+ T cells with further subsetting (e.g., CD8+, Va7.2+, etc.). For yeast, selection was gated on Cmyc+ or HA+ populations using UltraComp eBeads (eBiosciences) for compensation.

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.